

MSc Mathematics
Track: Data Science and Optimisation

Master thesis

Which Queue to Join?
**Routing Strategies in an Observable and Unobservable
Queueing Network**

by

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February 12, 2025

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I would like to dedicate this thesis to my mother Lidija, whose hard work and determination made this possible.

Abstract

This thesis examines strategic routing decisions in a feed-forward queueing network where customers choose which queues to visit based on available information. We analyse two scenarios: an unobservable setting, where customers know only the rate of arrival and rates of service at each queue, and a fully observable setting, where customers also know queue lengths. In the unobservable setting, we derive the equations and inequalities which characterise the Nash equilibrium and the social optimum, and numerically obtain these two strategies. Then, a numerical analysis of the solutions is done. In the observable setting, an algorithm is presented which finds an approximation to the Nash equilibrium. The resulting equilibrium is studied numerically, and compared to simpler strategies in terms of how much social cost it incurs.

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1. Introduction

Waiting in a queue is universally regarded as a waste of time. Standing in line, be it at a bank or at an airport, means losing time that could be spent in a more meaningful way than counting the tiles on the ceiling. The outlook is even more gloomy when we consider that sometimes, we have to wait not only in one queue, but must navigate through a network of queues. This is why queueing theorists spend much of their time quantifying how the design of a queue impacts the average time a customer spends in the system, or how long they will have to wait before obtaining service. The hope is to obtain insights which could then be relayed to the system designer, who could make changes to reduce waiting times and increase customer satisfaction and/or the system profits.

An important assumption that often simplifies analysis is that customers join the system independently of each other, and all join in a similar fashion. This assumption is not always realistic though. Consider for a moment a person on their way to work who has just spotted an outdoor coffee stand. They would like to get some coffee, but they also cannot be late to work. If the line for coffee is somewhat short, they will join it. If there are many customers, they will estimate that their waiting time will be too long, and they will skip the coffee. This means that the joining behaviour of the customer depends on the joining behaviour of those customers that are already in the queue, and those that considered to join the queue, but decided not to. If we wanted to describe the behaviour of this system, we could not rely on classical results from queueing theory. Instead, we would turn to strategic queueing theory, a field that lies at the intersection of classical queueing theory and game theory. Indeed, customers observing the state of the system and then deciding whether to join is very similar to players observing the state of the game and deciding which move to make next.

In this paper, we will not consider a single queue with a join/balk decision as the one described above. Such queues have already been studied extensively (see e.g. [14], [5], [6]). We will instead consider a feed-forward network where customers must decide how to route through this network. Such problems have gathered less attention, in part because studying queueing networks is generally much more difficult than studying single queue problems.

The network we study, shown in Figure 1.1, consists of five servers divided in two stages. Customers arrive according to a Poisson process, and all servers have exponentially distributed service times. An arriving customer must obtain service from exactly one server at each stage of the network, and they may choose which server to use in each stage. Depending on what servers were chosen in the previous stage, the customer may only have access to certain servers in the subsequent stage, as indicated in the figure by the arrows between servers. We first consider *selfish routing*: customers choose how to

route through the system in a way that minimises their individual cost, which we take to be their expected waiting time. There are possibly many different selfish strategies, each of which prescribes different actions, but all of which minimise the cost of an individual. We also consider *socially optimal routing*, where a central figure (social planner) directs customers through the system with the goal of minimising the social cost, which is some function of the costs of all customers in the system.

The strategies that customers use, selfish or socially optimal, depend on the cost function, i.e., the expected waiting time. The way that customers compute their expected waiting time depends in turn on how much information about the system is known at the time the decision is made. We consider two different scenarios: the unobservable and the (fully) observable case. In the former, the decision maker only knows how often customers arrive to the system and what the rate of service is at each server in the network. In the latter, the decision maker also knows exactly how long the line at each queue is at the time they need to make a decision. In the unobservable setting, we derive a selfish and a socially optimal strategy. We also study the so-called price of anarchy (PoA) [16] in this setting. In the observable setting, we approximate a selfish routing strategy and study its properties.

To conclude this chapter, we note that the overarching idea is not only to describe what happens in this particular network, but also to gain some insight into how customers behave in response to information in general feed-forward networks. However, we can imagine that our particular network can be useful to model a real-life situation as well. It could, for example, be used to represent the security and passport control queues in an airport. Depending on which security lane travellers use, they may only have access to certain passport control lanes.

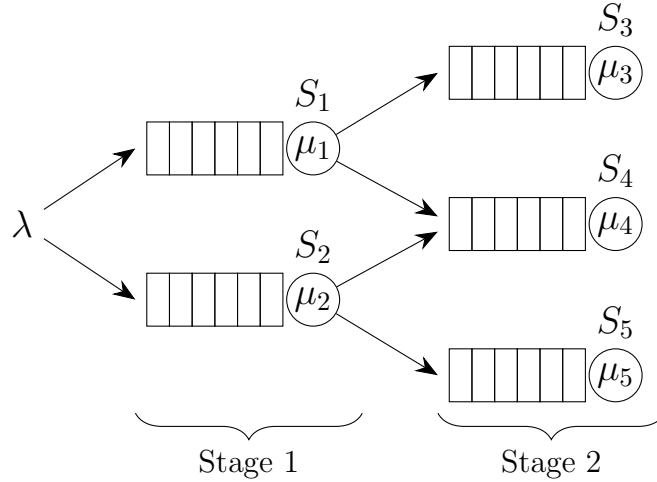


Figure 1.1: The network.

2. Literature Review

The work in strategic queueing was started by Naor in 1969. He assumed that customers arrive according to a Poisson process to a first-come-first-served exponential server, and they must decide upon entering the system whether they will join or balk. The customers incur a cost of C for each time unit spent in the system and a reward of R upon their service. Naor [14] derived the individual's optimal joining strategy, and the joining strategy which optimises social welfare, and concluded these are not necessarily equal to each other.

Since Naor, quite a few questions in strategic queueing have been answered. His model was expanded to include priorities [5], server breakdowns [6], batch service [4], etc. Other customer decisions were also considered, for example, the paper by Glazer and Hassin [7] allows customers to choose their time of arrival to a facility that is open during a fixed time interval. Assaf and Haviv [3] consider a model with processor sharing where multiple customers may use the server at the same time, but their utility decreases with the number of customers simultaneously using the server. In their paper, the customer's decision is not whether or not they will join, but if and when they will renege from the system, that is, leave it for good. For an overview of single server problems in strategic queueing, see Ibrahim [10]. Going further we will focus on routing in networks, as this is the topic of our research.

Strategic routing has received limited attention within the field of strategic queueing. However, it has been explored extensively in the context of algorithmic game theory, with key findings summarised in [16]. Most studies adopting an algorithmic game theory perspective on strategic routing assume everywhere continuous cost functions. The cost function in our work is the expected waiting time, seen as a function of the arrival rate. This function is not continuous on its whole domain. Therefore, most such results do not apply directly to our work. However, cost considerations specific to queueing theory are addressed in [15]. The author assumes a general network topology and takes as the cost function of a queue the expected waiting time in an M/M/1 queue. He then presents a bound for the price of anarchy in terms of how much extra capacity the queues in the network have.

As for routing problems in strategic queueing theory, some of the earliest work is done by Haight [9]. He studied the stationary state probabilities of a network with two identical parallel queues, under the JSQ (join the shortest queue) strategy. However, he did not consider different strategies or prove that this one is an equilibrium, so this paper may not be considered as one on strategic queueing. We will now present some papers that cover strategic routing in queueing networks. In all of these papers the topology of the network is different than ours, and unless otherwise specified, the routing decisions are made only once, upon arrival.

Arlotto et al. [2] studied a tandem of two stations, where customers must be serviced by both stations, but can choose the sequence in which they visit the stations. They proved the surprising result that under certain conditions all customers choose the same route through the system. The two biggest differences with our work are that they assume all the customers are present at the start of the game and that the network they study is not feed-forward.

Afimeimounga et al. [1] studied equilibrium routing strategies in two queues in parallel, where the first one is $M/1$ single-server infinite capacity queue, and the second one a $G(N)/\infty$ batch service queue. There is a dedicated stream of Poisson arrivals to batch service queue, and another general stream of Poisson arrivals, where customers choose which queue to join upon arrival. They show that a unique equilibrium strategy exists and show various properties of the strategy. This paper shares some qualities with our work. For example, the expected utility of a customer depends on customers who have not yet joined the system; after all, each customer who chooses the batch queue must wait for their batch to be complete before service commences. A large difference with our work is that the use of batch service allows the authors to significantly reduce their state space.

Logothetis and Economou [13] worked on a simplified model of a transportation station. Customers arrive according to a Poisson process to the station which offers multiple transportation facilities, such as a bus and a train. Vehicles arrive according to the same renewal process and have a certain number of seats available. The number of seats available is a random variable that is identically distributed within each transport facility. The authors derive both the individual equilibrium strategy and the socially optimal strategy. A major difference with our model is that here there are two stochastic arrival processes: the customers (passengers) and the servers (transport). There is also a possibility that there is no seating available, in which case the customer leaves the system for good. This cannot happen in our model.

Ji et al. [11] consider a two-node tandem network with exponentially distributed service times at each node. Customers arrive according to a Poisson process and choose whether to join or balk. If they join, they may renege later. Should they complete service, they receive a reward of R , and for each time unit spent in the system, they incur a cost of C . What makes this paper similar to ours is the fact that customers make the join/balk decision at each successive node in the tandem. The authors compare the equilibrium strategies across two information levels: the fully observable case and the partially observable case in which customers can only see the length of the queue in front of them. They find that, in most cases, the partially observable model gives higher throughput, but lower social welfare. They also find that customers never renege in practice. This is one of the few papers we found that concerns sequential decision-making, and does so across multiple information levels. The difference with our paper is, of course, the decision that customers make. Because all customers choose to leave the system if their utility is low enough, the state space is bounded from above by a certain threshold. Our paper does not have that property: if the servers are slow compared to the arrival process the system becomes congested and the number of customers at each queue may go to infinity.

3. Model

In this chapter, we will formalise the model. We study the queueing network shown in Figure 1.1. The network consists of five stations; each station consists of a server (labelled S_i) and its queue. There are two stages in the network: the first stage $\{S_1, S_2\}$ and the second stage $\{S_3, S_4, S_5\}$. All servers follow a “first-in-first-out” queueing discipline. The service time distribution of server i is exponential with rate μ_i . Customers arrive according to a Poisson process with rate λ . Throughout the text we assume $\mu_i > 0, \forall i \in \{1, 2, 3, 4, 5\}$ and that $\lambda < \mu_1 + \mu_2$ and $\lambda < \mu_3 + \mu_4 + \mu_5$. This is a necessary (but not sufficient) condition for stability. We assume the system is in steady state. We denote the expected waiting time at server S_i by $\mathbb{E}(W_i)$.

Customers will make decisions sequentially: once they join the network, they will choose whether to use server S_1 or S_2 , and then once they obtain this service, they will choose which of the two second stage servers available to them they will use. As the network figure suggests, if a customer used S_1 , she cannot use the server S_5 for her second service, and if she used S_2 , she cannot use S_3 for her second service. Upon service completion at the second stage server, all customers leave the system.

At the time of each decision, customers obtain some information about the network. In the unobservable case, customers know only their current location in the network and the network parameters $\lambda, \mu_1, \mu_2, \mu_3, \mu_4$, and μ_5 . In the fully observable case, customers also know the number of customers that are in each station, denoted by $(n_1, n_2, n_3, n_4, n_5)$. The state space $\mathcal{X}$ contains all the states of the network that a customer can perceive. In the unobservable case, there are multiple ways to define the state space. In both definitions we use, states never carry any information about the congestion of the system. In the fully observable case, the state space is $\mathbb{N}_0^5 \times \{0, 1, 2\}$. The first set contains information on how many customers are in the five queues. The second set contains the possible locations l of the tagged customer: $l = 0$ means the customer has just arrived to the network, $l = 1$ means she has just obtained service at S_1 , and $l = 2$ means she has just obtained service at S_2 .

The actions available to customers depend on her location in the network. We have the following action sets: $\mathcal{A}_0 = \{1, 2\}$, $\mathcal{A}_1 = \{3, 4\}$, and $\mathcal{A}_2 = \{4, 5\}$. Here action i means “join server S_i ”. The decision can be made deterministically, so that in each state we prescribe a single action. This is known in game theory as a *pure strategy*. If the decision is made stochastically, so that a customer goes to one server with probability $p \in (0, 1)$ or to the other server with probability $1 - p$, then the customer is using a *mixed strategy*. More formally, a strategy is a function from the state space $\mathcal{X}$ to the respective action space.

When forming strategies, customers will make use of the *cost function*, which is defined as their expected remaining waiting time in the system. We add the word remaining

here because when customers are making their second decision, they do not take into account the time they spent waiting for their first service. The goal is to find a particular strategy known as the Nash equilibrium, which is defined as follows.

Definition 3.1. Let $F(a, b)$ be the cost function of a customer who uses strategy a , when all other customers use strategy b . A symmetric *Nash* equilibrium is a strategy s_N such that

$$s_N \in \arg \min_{s \in \mathcal{S}} F(s, s_N).$$

In other words, a strategy s_N is a (symmetric Nash) equilibrium if it is one of the best responses against itself. The prefix symmetric means that we assume that all customers are applying the same strategy. This is a common assumption, as it simplifies the analysis and is realistic in real-life scenarios.

Aside from finding individually optimal strategies, often of interest is a strategy that minimises the social cost. The exact definition of the social cost depends on the specifics of the problem, but it always represents some aggregate cost of all players. Regardless of the exact definition of the social cost, we can define a socially optimal strategy.

Definition 3.2. Let $SC(s)$ be the social cost of the system, given that all customers use strategy s . The socially optimal strategy, also called the *social optimum* is a strategy s_S such that

$$s_S = \arg \min_{s \in \mathcal{S}} SC(s).$$

These two strategies do not necessarily coincide, especially when the choices of one customer impose negative externalities on other customers, which is certainly the case in our model. For example, a customer joining queue 1 increases the expected waiting time of a customer who will join this queue after them.

The difference between Nash routing and socially optimal routing can be summarised through the previously mentioned price of anarchy and price of stability, defined below.

Definition 3.3. The price of anarchy (PoA) and the price of stability (PoS) are defined as follows

$$PoA = \frac{\max_{s_N} SC(s_N)}{SC(s_S)},$$

$$PoS = \frac{\min_{s_N} SC(s_N)}{SC(s_S)}.$$

In words, the price of anarchy is the quotient of the maximal social cost in a Nash equilibrium, and the social cost in a social optimum. The price of stability is obtained by considering the Nash equilibrium with the smallest social cost, rather than the one with largest social cost. If the Nash equilibrium is unique, they coincide.

4. Unobservable Case - Analytical Solutions

In this chapter we focus on the unobservable case, where customers only know the network parameters $\lambda, \mu_1, \dots, \mu_5$. After formalising the problem, we show how to obtain the Nash equilibrium strategy in Section 4.1. We formalise the minimisation problem which defines the social optimum in Section 4.2.

We introduce the state and action space in two different, but equivalent ways. The first representation of the problem is one with distinct states in the system. We include this representation because it generalises nicely to the unobservable case, which we will study in Chapter 6. We can define the state space to be $\mathcal{X} = \{0, 1, 2\}$, where 0 indicates that the customer making the decision has just arrived to the system, 1 indicates that she has just obtained service at server S_1 , and 2 indicates that she has just obtained service at server S_3 . Denoting the action space at state i as $\mathcal{A}_i$, we have $\mathcal{A}_0 = \{1, 2\}$, $\mathcal{A}_1 = \{3, 4\}$ and $\mathcal{A}_2 = \{4, 5\}$ as before. Now a strategy is a function s , such that $s(0) = (p, 1 - p)$, $s(1) = (q, 1 - q)$, and $s(2) = (r, 1 - r)$, where p, q , and r are the probabilities of choosing the first action in the respective action space. We can obtain pure strategies by setting p, q and r to some combination of zeros and ones. We have the following characterisation of a Nash equilibrium strategy.

Definition 4.1. We say a strategy (p_N, q_N, r_N) is a Nash equilibrium if the following holds. If all customers follow the Nash strategy (p_N, q_N, r_N) and a new customer joins, then the new customer's expected waiting time in the system is minimised precisely when she also uses the strategy (p_N, q_N, r_N) .

We now present another description of the problem, one that is closer to selfish routing problems in algorithmic game theory. In this representation, there are no states (as is generally the case with unobservable strategic queueing problems). As we described in Chapter 3, customers make two decisions, once upon joining the system and once after obtaining their first service. However, because customers will have exactly the same state information when they are making their second decision, they can decide on their second stage server as soon as they join the system. This means that, in the unobservable case, we can define the action space to be the set of all routes $\mathcal{A} = \{R_1, R_2, R_3, R_4\}$. Here R_1 denotes the event of using the route $\{S_1, S_3\}$, R_2 means using route $\{S_1, S_4\}$, R_3 means using route $\{S_2, S_4\}$, and R_4 means using route $\{S_2, S_5\}$. We will sometimes use R_i to mean simply route i , rather than the event “use route i ”. The costs are once again the customer's expected waiting time in the system. A strategy s is of the form (p_1, p_2, p_3, p_4) , where p_i is the probability of following route R_i . Pure strategies are obtained by setting one p_i to 1, and the rest to 0.

We can relate the two interpretations of the problem as follows. If we have the strategy (p_1, p_2, p_3, p_4) and we would like to know the corresponding p, q , and r , we can compute

$$\begin{aligned} p &= \mathbb{P}(\text{"use } S_1\text{"}) = \mathbb{P}(R_1 \cup R_2) = p_1 + p_2, \\ q &= \mathbb{P}(\text{"use } S_3\text{"} | \text{"use } S_1\text{"}) = \frac{p_1}{p_1 + p_2}, \\ r &= \mathbb{P}(\text{"use } S_4\text{"} | \text{"use } S_2\text{"}) = \frac{p_3}{p_3 + p_4}. \end{aligned}$$

We can just as easily map (p, q, r) back to (p_1, p_2, p_3, p_4) , so that we can switch representations when convenient. The first representation is more natural given that customers make decision sequentially, and it shows more clearly what happens when we change the network parameters. On the other hand, we will see that it is often useful to assume that customers make only one decision, as it simplifies the qualitative arguments about the Nash equilibrium.

Any strategy (p, q, r) induces a Jackson network, shown in Figure 4.1. This fact will be used when we derive the Nash equilibrium.

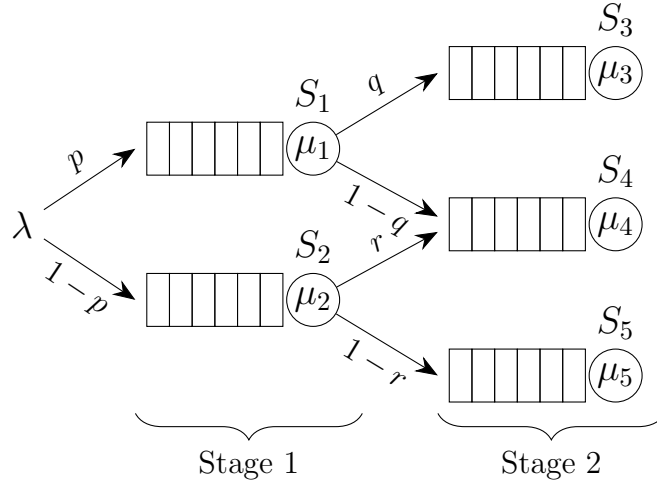


Figure 4.1: The network in the unobservable case.

In what follows we present some analytical solutions. More precisely, we show how to compute the Nash equilibrium routing probabilities, present the closed form solutions for some simple cases, and show how to obtain the social optimum.

4.1. Nash Equilibrium

We begin this section with a result that bounds the number of deterministic choices in a Nash equilibrium to at most one.

Lemma 4.2. *Suppose $s = (p, q, r)$ where at least two coordinates of s are either 0 or 1. Then s is not a Nash equilibrium.*

Proof. We have two cases, one where all three coordinates are either 0 or 1, and one where two coordinates are either 0 or 1.

Case 1. All three coordinates of s are either 0 or 1.

If this is the case, s prescribes both the first and second stage server deterministically, and we can identify it with one of the four possible routes. Suppose every customer follows this pure strategy. Then the expected waiting time of a joining customer, if following the same strategy, is non-zero. The expected waiting time if following a route which consists of the two servers that are not in s is 0. A simple check shows that such a strategy s' exists for any pure strategy s . Then s' is strictly a better response to s than s is to itself. By definition, s cannot be a Nash equilibrium.

Case 2. Exactly one coordinate of s is strictly between 0 and 1.

Subcase (i): $p = 0$ or $p = 1$

First suppose $p = 0$. Then the expected waiting time on one of the routes R_3 and R_4 is strictly positive. On the other hand, the expected waiting time on route R_1 is 0. It is therefore of interest for a tagged customer to deviate from the strategy and use strategy $(1, 1, 0)$, i.e. use route R_1 . Again, the strategy is not a best response against itself, and therefore also not a Nash equilibrium. A similar argument shows that p cannot be 1.

Subcase (ii): $p \in (0, 1)$

There are four combinations left: $(p, 1, 1)$, $(p, 0, 1)$, $(p, 1, 0)$ and $(p, 0, 0)$. We will explain why $(p, 1, 1)$ cannot be a Nash equilibrium. The proof for the other three combinations is very similar. If $s = (p, 1, 1)$, the expected waiting time at server S_5 is 0, while the expected waiting time at server S_4 is non-zero. We have

$$\begin{aligned}\mathbb{E}(W|(p, 1, 1)) &= p\mathbb{E}(W_1) + p\mathbb{E}(W_3) + (1-p)\mathbb{E}(W_2) + (1-p)\mathbb{E}(W_4) \\ &> p\mathbb{E}(W_1) + p\mathbb{E}(W_3) + (1-p)\mathbb{E}(W_2) + (1-p)\mathbb{E}(W_5) = \mathbb{E}(W|(p, 1, 0)).\end{aligned}$$

It is again of interest to the joining customer to use a different strategy, so $(p, 1, 1)$ is not a Nash equilibrium. □

We continue with our analysis of the Nash equilibrium. We have the following effective arrival rates to each queue

$$\begin{aligned}\lambda_1 &= p\lambda, & \lambda_2 &= (1-p)\lambda, & \lambda_3 &= pq\lambda, \\ \lambda_4 &= (p(1-q) + r(1-p))\lambda, & \lambda_5 &= (1-r)(1-p)\lambda,\end{aligned}$$

and the following node loads

$$\begin{aligned}\rho_1 &= \frac{p\lambda}{\mu_1}, & \rho_2 &= \frac{(1-p)\lambda}{\mu_2}, & \rho_3 &= \frac{pq\lambda}{\mu_3}, \\ \rho_4 &= \frac{(p(1-q) + r(1-p))\lambda}{\mu_4}, & \rho_5 &= \frac{(1-r)(1-p)\lambda}{\mu_5}.\end{aligned}$$

Since we have a Jackson network, we can compute the expected waiting times using the formula for the M/M/1 queue with arrival rate λ and service rate μ . The formula for the expected waiting time is then $\mathbb{E}(W) = \frac{\rho}{\mu(1-\rho)} = \frac{\lambda}{\mu(\mu-\lambda)}$.

We get the following expressions

$$\begin{aligned}\mathbb{E}(W_1) &= \frac{p\lambda}{\mu_1(\mu_1 - p\lambda)}, \\ \mathbb{E}(W_2) &= \frac{(1-p)\lambda}{\mu_2(\mu_2 - (1-p)\lambda)}, \\ \mathbb{E}(W_3) &= \frac{pq\lambda}{\mu_3(\mu_3 - pq\lambda)}, \\ \mathbb{E}(W_4) &= \frac{(p(1-q) + r(1-p))\lambda}{\mu_4(\mu_4 - (p(1-q) + r(1-p))\lambda)}, \\ \mathbb{E}(W_5) &= \frac{(1-p)(1-r)\lambda}{\mu_5(\mu_5 - (1-p)(1-r)\lambda)}.\end{aligned}$$

The following lemma characterises the Nash equilibrium; it states that in a Nash equilibrium, all used routes must have equal expected waiting time, and the expected waiting time of any unused route must be strictly larger than of those routes used in equilibrium. If this was not the case, a joining customer would have an incentive to use the route with the lowest waiting time, and would therefore deviate from the strategy.

Lemma 4.3. *Let s be a strategy. Let $\mathcal{R}_s$ be the set of all routes that are used with non-zero probability under this strategy, so that $\sum_{R_i \in \mathcal{R}_s} \mathbb{P}_s(R_i) = 1$, where $\mathbb{P}_s(R_i)$ is the probability of using route R_i under strategy s . Let $\mathbb{E}(W_R|s)$ denote the expected waiting time of the tagged customer if she follows route R , while all other customers follow the route prescribed by strategy s . If s is a Nash equilibrium, this quantity is the same for all routes R in $\mathcal{R}_s$. Furthermore, if s is a Nash equilibrium and a route R is not in $\mathcal{R}_s$, this route cannot have a lower expected waiting time $\mathbb{E}(W_R|s)$ than the routes in $\mathcal{R}_s$.*

Conversely, if s is a strategy such that $\mathbb{E}(W_R|s)$ is equal to some constant w for all routes in $\mathcal{R}_s$ and $\mathbb{E}(W_{R^}|s) \geq w$ for routes R^* that are not in $\mathcal{R}_s$, then s is a Nash equilibrium.*

Proof. Throughout the proof we use $\mathbb{E}(W_{\tilde{s}}|s)$ to denote the expected waiting time of a tagged customer who uses strategy $\tilde{s}$ when all other customers use strategy s . We first show that if $\mathbb{E}(W_R|s)$ is not equal for all routes in $\mathcal{R}_s$, then s is not a Nash equilibrium. Indeed, if this is the case then there is a route R_* in $\mathcal{R}_s$ such that its expected waiting time in response to s is the lowest of all routes in $\mathcal{R}_s$. Let s_* be the pure strategy of a tagged customer which consists of using this route with probability 1. We have

$$\mathbb{E}(W_{s_*}|s) = 1 \cdot \mathbb{E}(W_{R_*}|s) = \sum_{R_i \in \mathcal{R}_s} \mathbb{P}_s(R_i) \mathbb{E}(W_{R_*}|s) < \sum_{R_i \in \mathcal{R}_s} \mathbb{P}_s(R_i) \mathbb{E}(W_{R_i}|s) = \mathbb{E}(W_s|s).$$

We see that s is not a best response against itself, and therefore not a Nash equilibrium.

Next, we show that if a route is not in $\mathcal{R}_s$, it cannot have a strictly lower expected waiting time as a response to s than the routes in $\mathcal{R}_s$. To this end, let R_* be a route

which is not in $\mathcal{R}_s$, and has a strictly lower expected waiting time than the routes in $\mathcal{R}_s$. Let R be any route in $\mathcal{R}_s$. Let s_* be the strategy that agrees with s on all routes except R and R_* ; it gives probability 0 to R , and probability $\mathbb{P}_s(R)$ to R_* . Then

$$\begin{aligned}\mathbb{E}(W_s|s) &= \sum_{R_i \in \mathcal{R}_s} \mathbb{P}_s(R_i) \mathbb{E}(W_{R_i}|s) \\ &= \sum_{R_i \in \mathcal{R}_s \setminus \{R\}} \mathbb{P}_s(R_i) \mathbb{E}(W_{R_i}|s) + \mathbb{P}_s(R) \mathbb{E}(W_R|s) \\ &> \sum_{R_i \in \mathcal{R}_s \setminus \{R\}} \mathbb{P}_s(R_i) \mathbb{E}(W_{R_i}|s) + \mathbb{P}_s(R) \mathbb{E}(W_{R_*}|s) = \mathbb{E}(W_s|s_*).\end{aligned}$$

Once again, we see s is not a Nash equilibrium.

Finally, we show that if $\mathbb{E}(W_R|s)$ is equal for all routes R in $\mathcal{R}_s$ and $\mathbb{E}(W_R|s)$ is at least as high for all routes not in $\mathcal{R}_s$, then s is a Nash equilibrium. Let $\mathbb{E}(W_R|s) = w$ for all routes in $\mathcal{R}_s$. Then

$$\mathbb{E}(W_s|s) = \sum_{R_i \in \mathcal{R}_s} \mathbb{P}_s(R_i) \mathbb{E}(W_{R_i}|s) = w \sum_{R_i \in \mathcal{R}_s} \mathbb{P}_s(R_i) = w.$$

Now let s_* be any other strategy, with its set of used routes $\mathcal{R}_{s_*}$. We have

$$\mathbb{E}(W_{s_*}|s) = \sum_{R_i \in \mathcal{R}_{s_*}} \mathbb{P}_{s_*}(R_i) \mathbb{E}(W_{R_i}|s) = \sum_{R_i \in \mathcal{R}_{s_*} \cap \mathcal{R}_s} \mathbb{P}_{s_*}(R_i) w + \sum_{R_i \in \mathcal{R}_{s_*} \setminus \mathcal{R}_s} \mathbb{P}_{s_*}(R_i) \mathbb{E}(W_{R_i}|s).$$

By assumption, it holds that $\mathbb{E}(W_{R_i}|s) \geq w$ for routes R_i in $\mathcal{R}_{s_*} \setminus \mathcal{R}_s$. Thus,

$$\begin{aligned}\mathbb{E}(W_{s_*}|s) &= \sum_{R_i \in \mathcal{R}_{s_*} \cap \mathcal{R}_s} \mathbb{P}_{s_*}(R_i) w + \sum_{R_i \in \mathcal{R}_{s_*} \setminus \mathcal{R}_s} \mathbb{P}_{s_*}(R_i) \mathbb{E}(W_{R_i}|s) \\ &\geq \sum_{R_i \in \mathcal{R}_{s_*} \cap \mathcal{R}_s} \mathbb{P}_{s_*}(R_i) w + \sum_{R_i \in \mathcal{R}_{s_*} \setminus \mathcal{R}_s} \mathbb{P}_{s_*}(R_i) w \\ &= \sum_{R_i \in \mathcal{R}_{s_*}} \mathbb{P}_{s_*}(R_i) w = w.\end{aligned}$$

Putting everything together, we see that $\mathbb{E}(W_{s_*}|s) \geq \mathbb{E}(W_s|s)$ for any strategy $s_* \neq s$, and therefore s is a Nash equilibrium. $\square$

We call an equilibrium a *zero routing* equilibrium if not all routes are used. If all routes are used, we call Nash a *non-zero routing* equilibrium. The latter is the topic of the following subsection.

4.1.1. The non-zero routing case

In this subsection we assume all routes are used, so $p \notin \{0, 1\}, r \notin \{0, 1\}, q \notin \{0, 1\}$. We have the following characterisation of a Nash equilibrium.

Lemma 4.4. *A strategy (p, q, r) is a non-zero routing Nash equilibrium if and only if it is a solution to the following set of equations.*

$$p\mu_2(\mu_2 - (1 - p)\lambda) = (1 - p)\mu_1(\mu_1 - p\lambda), \quad (4.1)$$

$$pq\mu_4(\mu_4 - (p(1 - q) + r(1 - p))\lambda) = (p(1 - q) + r(1 - p))\mu_3(\mu_3 - pq\lambda), \quad (4.2)$$

$$(p(1 - q) + r(1 - p))\mu_5(\mu_5 - (1 - p)(1 - r)\lambda) = (1 - p)(1 - r)\mu_4(\mu_4 - (p(1 - q) + r(1 - p))\lambda). \quad (4.3)$$

Proof. We use the fact that a strategy is an equilibrium if and only if the expected waiting time of a tagged customer is equal over all routes used and greater or equal in routes not used, as proven in Lemma 4.3.

Furthermore, since the waiting time of a tagged customer on a route is simply equal to the sum of the waiting times at each node visited, we get the following expression

$$\mathbb{E}(W_1) + \mathbb{E}(W_3) = \mathbb{E}(W_1) + \mathbb{E}(W_4) = \mathbb{E}(W_2) + \mathbb{E}(W_4) = \mathbb{E}(W_2) + \mathbb{E}(W_5).$$

After subtracting equal terms on each side, we obtain the following equations

$$\begin{aligned} \mathbb{E}(W_1) &= \mathbb{E}(W_2), \\ \mathbb{E}(W_3) &= \mathbb{E}(W_4) = \mathbb{E}(W_5). \end{aligned}$$

Plugging in the formulas for the expected waiting times gives the following equations

$$\begin{aligned} \frac{p\lambda}{\mu_1(\mu_1 - p\lambda)} &= \frac{(1 - p)\lambda}{\mu_2(\mu_2 - (1 - p)\lambda)}, \\ \frac{pq\lambda}{\mu_3(\mu_3 - pq\lambda)} &= \frac{(p(1 - q) + r(1 - p))\lambda}{\mu_4(\mu_4 - (p(1 - q) + r(1 - p))\lambda)}, \\ \frac{(p(1 - q) + r(1 - p))\lambda}{\mu_4(\mu_4 - (p(1 - q) + r(1 - p))\lambda)} &= \frac{(1 - p)(1 - r)\lambda}{\mu_5(\mu_5 - (1 - p)(1 - r)\lambda)}. \end{aligned}$$

Dividing each side of all three equations by λ and performing cross multiplication gives

$$\begin{aligned} p\mu_2(\mu_2 - (1 - p)\lambda) &= (1 - p)\mu_1(\mu_1 - p\lambda), \\ pq\mu_4(\mu_4 - (p(1 - q) + r(1 - p))\lambda) &= (p(1 - q) + r(1 - p))\mu_3(\mu_3 - pq\lambda), \\ (p(1 - q) + r(1 - p))\mu_5(\mu_5 - (1 - p)(1 - r)\lambda) &= (1 - p)(1 - r)\mu_4(\mu_4 - (p(1 - q) + r(1 - p))\lambda), \end{aligned}$$

which is the desired result. □

Equation (4.1) is a quadratic in p , and it rewrites to $Ap^2 + Bp + C = 0$ with

$$\begin{aligned} A &= (\mu_1 - \mu_2)\lambda, \\ B &= -(\mu_1 - \mu_2)\lambda - \mu_1^2 - \mu_2^2, \\ C &= \mu_1^2, \end{aligned}$$

and therefore has two solutions, $p_1 = \frac{-B-\sqrt{D}}{2A}$ and $p_2 = \frac{-B+\sqrt{D}}{2A}$ where $D = B^2 - 4AC$. It is certainly possible that for certain combinations of parameters p_1 or p_2 is not meaningful as a routing probability. In that case we say p_i is not feasible. We say p_i is feasible if it satisfies the following inequalities

1. $p \geq 0$,
2. $p \leq 1$,
3. $\lambda p < \mu_1$,
4. $\lambda(1 - p) < \mu_2$.

The following lemma shows that for all combinations of parameters, p_1 is feasible and p_2 is not. The proof works as follows: we define a function $f(p) = \mathbb{E}W_1(p) - \mathbb{E}W_2(p)$, so that if we find a zero of f , we have found a point p where $\mathbb{E}W_1(p) = \mathbb{E}W_2(p)$, as we would like. We then construct an interval I which contains all values p which are feasible. We apply the Intermediate value theorem to show that f has a zero in the interval I . The theorem states that if f is continuous on the interval $[a, b]$ and $f(a) < 0, f(b) > 0$, then f has a root in this interval. Finally we show that p_2 is not in the interval. This implies that p_1 is a zero of f in I , i.e., a feasible solution of Equation (4.1).

Lemma 4.5. *The only feasible solution to Equation (4.1) is $p_1 = \frac{-B-\sqrt{D}}{2A}$, in the sense that it is the only solution p that satisfies $p \geq 0$, $p \leq 1$, $\lambda p < \mu_1$ and $\lambda(1 - p) < \mu_2$.*

Proof. We will apply the Intermediate value theorem to the interval $I = [1 - \frac{\mu_2}{\lambda}, \frac{\mu_1}{\lambda}] \cap [0, 1]$. We first show that I is a well-defined, non-empty interval. Note that $[1 - \frac{\mu_2}{\lambda}, \frac{\mu_1}{\lambda}]$ is a well defined interval, since $1 - \frac{\mu_2}{\lambda} < \frac{\mu_1}{\lambda} \iff \lambda < \mu_1 + \mu_2$. The latter statement is a part of the stability conditions we impose in Section 3. So I is a well-defined interval. Also, the intersection of $[1 - \frac{\mu_2}{\lambda}, \frac{\mu_1}{\lambda}]$ and $[0, 1]$ is non-empty as $0 < \mu_1/\lambda$, so I is non-empty.

The functions $\mathbb{E}W_1(p)$ and $\mathbb{E}W_2(p)$ are continuous in I since they are continuous at any interior point of the interval, and at the end points we have

$$\begin{aligned} \lim_{p \rightarrow 1 - \frac{\mu_2}{\lambda} +} \mathbb{E}W_2(p) &= \infty = \mathbb{E}W_2\left(1 - \frac{\mu_2}{\lambda}\right), \\ \lim_{p \rightarrow \frac{\mu_1}{\lambda} -} \mathbb{E}W_1(p) &= \infty = \mathbb{E}W_1\left(\frac{\mu_1}{\lambda}\right). \end{aligned}$$

As $\mathbb{E}W_1(p)$ and $\mathbb{E}W_2(p)$ are continuous in the interval I , their linear combination $f(p)$ is too.

To show the function changes signs on the interval, note that $f(1 - \frac{\mu_2}{\lambda}) = \mathbb{E}W_1(1 - \frac{\mu_2}{\lambda}) - \mathbb{E}W_2(1 - \frac{\mu_2}{\lambda}) = \mathbb{E}W_1(1 - \frac{\mu_2}{\lambda}) - \infty < 0$, whatever finite value $\mathbb{E}W_1(1 - \frac{\mu_2}{\lambda})$ gives. Similarly $f(\frac{\mu_1}{\lambda}) = \mathbb{E}W_1(\frac{\mu_1}{\lambda}) - \mathbb{E}W_2(\frac{\mu_1}{\lambda}) = \infty - \mathbb{E}W_2(\frac{\mu_1}{\lambda}) > 0$.

Therefore f has at least one root in I . Since any p in I satisfies the four feasibility inequalities listed above, the root in I is feasible too.

The roots of f are the solutions of Equation (4.1), by definition of f . Therefore, if $p_2 = \frac{-B+\sqrt{D}}{2A}$ is not in I , it follows that $\frac{-B-\sqrt{D}}{2A}$ is. We divide the proof that p_2 is not in $(1 - \frac{\mu_2}{\lambda}, \frac{\mu_1}{\lambda})$ into two cases.

Case 1. $\mu_1 < \mu_2$.

In this case $A < 0, C > 0$, so $4AC < 0$, and therefore $B^2 - 4AC > B^2$. Squaring both sides gives $\sqrt{D} > B$. This implies that $-B + \sqrt{D}$, the numerator of p_2 , is positive. Therefore $p_2 = \frac{-B+\sqrt{D}}{2A}$ is a quotient of a positive and a negative number, and is therefore negative and unfeasible.

Case 2. $\mu_1 > \mu_2$. To prove this case, we will use Vieta's formulas to show that $p_1 p_2 > 1$. We will then show that $p_1 < 1$ which implies that $p_2 > 1$ and therefore p_2 is unfeasible.

In this case it holds that $A > 0, C > 0, B < 0$. By Vieta's formulas we have

$$p_1 \cdot p_2 = C/A = \frac{\mu_1^2}{(\mu_1 - \mu_2)\lambda} > \frac{\mu_1^2}{(\mu_1 - \mu_2)(\mu_1 + \mu_2)} = \frac{\mu_1^2}{\mu_1^2 - \mu_2^2} > 1,$$

where the first inequality follows from the fact that $\lambda < \mu_1 + \mu_2$, and the second one from the fact that both μ_1^2 and $\mu_1^2 - \mu_2^2$ are positive, and $\mu_1^2 > \mu_1^2 - \mu_2^2$.

It holds by definition of p_1 that $p_1 < 1$ if and only if

$$\sqrt{D} > -B - 2A. \quad (4.4)$$

We can easily see that the RHS of Inequality (4.4) is positive

$$\begin{aligned} -B - 2A &= \mu_1^2 + \mu_2^2 - \lambda(\mu_1 - \mu_2) \\ &> \mu_1^2 + \mu_2^2 - (\mu_1 + \mu_2)(\mu_1 - \mu_2) \\ &= 2\mu_2^2 > 0. \end{aligned}$$

The first inequality follows from $\lambda < \mu_1 + \mu_2$. As the RHS of Inequality (4.4) is positive, we can square both sides of the inequality to see

$$(\sqrt{B^2 - 4AC})^2 > (-B - 2A)^2 \iff -4AC > 4AB + 4A^2 \iff A + B + C < 0.$$

Plugging in the expressions for A, B, C gives

$$A + B + C = (\mu_1 - \mu_2)\lambda - (\mu_1 - \mu_2)\lambda - \mu_1^2 - \mu_2^2 + \mu_1^2 = -\mu_2^2 < 0,$$

so $p_1 < 1$. Since $p_1 p_2 > 1$ and $p_1 < 1$, it follows that p_2 is greater than 1, and therefore infeasible.

It follows that $p_1 = \frac{-B-\sqrt{D}}{2A}$ is the only feasible solution to Equation (4.1). $\square$

That completes the analysis of Equation (4.1), and we can treat p as a constant in $(0, 1)$ in our further analysis of Equations (4.2) and (4.3).

First, we can isolate r from Equation (4.2) as this equation is linear in r . The equation rewrites to

$$\begin{aligned} qp\mu_4(\mu_4 - p\lambda + pq\lambda - r\lambda + rp\lambda) &= p\mu_3^2 - \mu_3qp^2\lambda - pq\mu_3^2 + \mu_3q^2p^2\lambda \\ &\quad + r\mu_3^2 - \mu_3qpr\lambda - rp\mu_3^2 + \mu_3rqp^2\lambda. \end{aligned}$$

Collecting all terms with r on one side, we obtain

$$\begin{aligned} r(p^2\lambda q\mu_4 - p\lambda q\mu_4 - \mu_3^2 + \mu_3qp\lambda + p\mu_3^2 - \mu_3qp^2\lambda) &= p\mu_3^2 - \mu_3qp^2\lambda - pq\mu_3^2 \\ &\quad + \mu_3q^2p^2\lambda - qp\mu_4^2 + qp^2\lambda\mu_4 - q^2p^2\lambda\mu_4. \end{aligned}$$

We therefore have the following expression for r

$$r = \frac{p\mu_3^2 - \mu_3qp^2\lambda - pq\mu_3^2 + \mu_3q^2p^2\lambda - qp\mu_4^2 + qp^2\lambda\mu_4 - q^2p^2\lambda\mu_4}{p^2\lambda q\mu_4 - p\lambda q\mu_4 - \mu_3^2 + \mu_3qp\lambda + p\mu_3^2 - \mu_3qp^2\lambda}. \quad (4.5)$$

We can now square r and plug in the expressions for r and r^2 into Equation (4.3). We obtain a polynomial equation $P(q) = 0$ of degree 4 in q , where $P(q) = C_0 + C_1q + C_2q^2 + C_3q^3 + C_4q^4$. The coefficients can be found in Table C.1 in the appendix. The analytical solution we obtain is problematic in some ways: the formula is too complicated to get any insights from, and we do not know the parameter values for which both r and q are in the domain $[0, 1]$. There is, however, one special case where we can get the closed form solution easily. The proof relies on the fact that when $\lambda = 1, \mu_1 = \mu_2$, and $\mu_3 = \mu_4$, then the fourth degree polynomial discussed above simplifies into a quadratic. This allows us to solve for q explicitly and show when the solution is feasible. Note that the polynomial also simplifies to a quadratic when $\lambda = 1, \mu_1 = \mu_2$, and $\mu_4 = \mu_5$, but we do not work out the solution.

Lemma 4.6. *If $\lambda = 1, \mu_1 = \mu_2, \mu_3 = \mu_4$, then the Nash equilibrium is $(p_N, q_N, r_N) = (\frac{1}{2}, q_*, 1 - q_*)$, where $q_* = \frac{-B + \sqrt{B^2 - 4AC}}{2A}$, $A = \mu_5 - \mu_3, B = \mu_3 - \mu_5 + \mu_5^2 + 2\mu_3^2$ and $C = -2\mu_3^2$.*

Proof. The function $P(q)$ is of the form $C_4q^4 + C_3q^3 + C_2q^2 + C_1q + C_0$, where each C_i is a linear combination of terms labelled $T_0, \dots, T_{17}$. Each T_i is in turn a function of $p, \lambda, \mu_1, \mu_2, \mu_3, \mu_4$ and μ_5 . If $\mu_1 = \mu_2$, then it holds that $p = \frac{1}{2}$. We compute the expressions T_i first. Since $\mu_3 = \mu_4$, each expression which contains $\mu_3 - \mu_4$ is equal to 0, so $T_1 = T_6 = T_9 = 0$. Also,

$$\begin{aligned} T_2 &= 2p^3\lambda(2\mu_3p\mu_4 + \mu_3\mu_4(\mu_3 - \mu_4) + \mu_4^3 - \mu_3^3 - p\lambda(\mu_3^2 + \mu_4^2)) \\ &= p(2\mu_3\mu_4 - \lambda(\mu_3^2 + \mu_4^2)) = 0. \end{aligned}$$

Similarly,

$$\begin{aligned} T_{13} &= p^2\lambda^2[\mu_4^2 + \mu_3^2 + (\mu_3p)^2 + (p\mu_4)^2 + 4\mu_3\mu_4p - 2\mu_3^2p - 2\mu_3\mu_4 - 2p^2\mu_3\mu_4 - 2p\mu_4^2] \\ &= p^2\lambda^2(p^2(\mu_3 - \mu_4)^2 + 2p(\mu_3 - \mu_4)^2 + (\mu_3 - \mu_4)^2) = 0. \end{aligned}$$

Finally,

$$T_{14} = 2p\lambda\mu_3^2[p^2\mu_4 + \mu_4 + 2\mu_3p - p^2\mu_3 - 2p\mu_4 - \mu_3] = 0.$$

Continuing with the computations, it is easy to confirm that

$$\begin{aligned} T_0 &= \frac{1}{4}(\mu_5 - \mu_3), & T_3 &= \mu_3^4, & T_4 &= -\mu_3^4, & T_5 &= \frac{1}{4}\mu_3^4, \\ T_7 &= -\mu_3^2, & T_8 &= \frac{1}{2}\mu_3^2, & T_{10} &= -\frac{1}{2}\mu_3^2, & T_{11} &= \frac{1}{2}(\mu_3^2 + \mu_5^2), \\ T_{12} &= -\frac{1}{4}(\mu_5 - \mu_3), & T_{15} &= \frac{1}{4}\mu_3^4, & T_{16} &= \frac{1}{4}(\mu_5 - \mu_3) - \frac{1}{2}\mu_5^2, \\ T_{17} &= \frac{1}{2}(\mu_5^2 - \mu_3^2) + \frac{1}{4}(\mu_3 - \mu_5). \end{aligned}$$

Plugging these into the expressions for C_i , we see that $C_3 = T_2C_0 + T_6T_9T_{11} + T_7T_9T_{12} + T_{13}T_{16} = 0$ and $C_4 = T_1T_0 + T_6T_9T_{12} = 0$ as each summand of these coefficients contains at least one zero T_i . For the rest of the coefficients, we have

$$C_0 = T_5T_0 + T_8T_{10}T_{11} + T_{15}T_{17} = -\frac{1}{4}\mu_3^6,$$

$$\begin{aligned} C_1 &= T_4T_0 + T_7T_{10}T_{11} + T_8T_9T_{11} + T_8T_{10}T_{12} + T_{14}T_{17} + T_{15}T_{16} \\ &= \frac{1}{4}\mu_3^4\left(\frac{1}{2}\mu_3 - \frac{1}{2}\mu_5 + \frac{1}{2}\mu_5^2 + \mu_3^2\right), \end{aligned}$$

$$\begin{aligned} C_2 &= T_3T_0 + T_6T_{10}T_{11} + T_7T_9T_{11} + T_7T_{10}T_{12} + T_8T_9T_{12} + T_{13}T_{17} + T_{14}T_{16} \\ &= \frac{1}{8}\mu_3^4(\mu_5 - \mu_3). \end{aligned}$$

Dividing every coefficient with $\mu_3^4/2$, it is clear we need to find the root of the polynomial $Aq^2 + Bq + C$, where $A = \mu_5 - \mu_3$, $B = \mu_3 - \mu_5 + \mu_5^2 + 2\mu_3^2$ and $C = -2\mu_3^2$.

In order for a point (p, q, r) to be Nash equilibrium, q must be a solution to $Ax^2 + Bx + C = 0$, and it must satisfy the following feasibility constraints

$$\begin{aligned} 0 &\leq q \leq 1, \\ pq\lambda &< \mu_3, \\ \lambda(p(1-q) + (1-p)r) &< \mu_4, \\ \lambda(1-p)(1-r) &< \mu_5. \end{aligned}$$

Plugging in $\lambda = 1, p = \frac{1}{2}$ and $\mu_3 = \mu_4$ into Expression (4.5), we can immediately verify

that $r_N = 2q_* - 1$. Therefore, the feasibility constraints simplify to

$$\begin{aligned} 0 &\leq q \leq 1, \\ \frac{1}{2}q &< \mu_3, \\ \frac{1}{2}(1-q) + \frac{1}{2}(2q-1) &< \mu_4 \iff \frac{1}{2}q < \mu_4, \\ \frac{1}{2}(1-r) &< \mu_5 \iff 1-q < \mu_5. \end{aligned}$$

The proof that $\frac{-B+\sqrt{B^2-4AC}}{2A}$ is feasible (and $\frac{-B-\sqrt{B^2-4AC}}{2A}$ is not) is very similar to the proof of Lemma 4.5 and therefore will not be included. Again, the idea is to split the proof into cases, manipulate the terms to show that $\frac{-B+\sqrt{B^2-4AC}}{2A}$ satisfies the feasibility conditions, and then use Vieta's formulas to show $\frac{-B-\sqrt{B^2-4AC}}{2A}$ does not. $\square$

Finally, we consider when a solution to the system of equations in Lemma 4.4 exists and when it is unique. To prove existence we make use of the Poincaré-Miranda theorem [12], a multi-variate equivalent to the Intermediate Value Theorem. The theorem follows.

Theorem 4.7. *Consider n continuous, real-valued functions of n variables, $f_1, \dots, f_n : [-1, 1]^n \mapsto \mathbb{R}$. Assume that for each variable x_i , the function f_i is non-positive when $x_i = -1$ and non-negative when $x_i = 1$. Then there is a point in the n -dimensional cube $[-1, 1]^n$ in which all functions are simultaneously equal to 0.*

To make use of the theorem above, we can define $f_1 : [0, 1] \mapsto \mathbb{R}$ as $f_1(q, r) = \mathbb{E}W_3(q, r) - \mathbb{E}W_4(q, r)$ and similarly $f_2 : [0, 1] \mapsto \mathbb{R}$ where $f_2(q, r) = \mathbb{E}W_4(q, r) - \mathbb{E}W_5(q, r)$. If we can find a zero (q_*, r_*) of $f = \begin{pmatrix} f_1 \\ f_2 \end{pmatrix}$, then we have found a Nash equilibrium, since then $\mathbb{E}W_3(q_*, r_*) = \mathbb{E}W_4(q_*, r_*)$ and $\mathbb{E}W_4(q_*, r_*) = \mathbb{E}W_5(q_*, r_*)$.

The theorem is not immediately applicable though; the issue is that the domain of f is not of the form $[-1, 1]^2$. This can be fixed easily. We define $g : [-1, 1]^2 \mapsto [0, 1]^2$ as $g((a, b)) = \frac{1}{2}(a+1, b+1)^T$, so that $F = f \circ g$ has the correct domain. We can then apply the theorem to F to find a zero of it. Then, a zero of F (i.e. (q_*, r_*) such that $f \circ g(q_*, r_*) = 0$) immediately gives a zero of f , namely $g(q_*, r_*)$.

The proof that F is continuous is given in Lemma A.1 and can be found in the appendix. The proof is somewhat tedious because we have to switch domains, but the main idea is to limit the domain of the function to those values where the expectations are continuous. The appendix also contains the proof that F_1 and F_2 satisfy the boundary conditions of Theorem 4.7, under some added conditions. Again, we have to be careful with the domain, but the main idea is to again restrict the domain.

Once the existence of a zero has been verified, we move on to proving injectivity, which will guarantee uniqueness of the zero. In the following lemma, we will show that $f = \begin{pmatrix} \mathbb{E}W_3(q, r) - \mathbb{E}W_4(q, r) \\ \mathbb{E}W_4(q, r) - \mathbb{E}W_5(q, r) \end{pmatrix}$ is injective on $[0, 1]^2$ under the same assumptions that guaranteed continuity. It will then follow that F is injective on $[-1, 1]^2$, since it is a composition of two injective functions, f and g .

Lemma 4.8. Assume $p\lambda < \mu_3$, $\lambda(1-p) < \mu_5$, and $\lambda < \mu_4$. The function $f : [0, 1]^2 \rightarrow \mathbb{R}^2$ given by $f = \begin{pmatrix} \mathbb{E}W_3(q, r) - \mathbb{E}W_4(q, r) \\ \mathbb{E}W_4(q, r) - \mathbb{E}W_5(q, r) \end{pmatrix}$ is injective on its domain.

Proof. We make a proof by contradiction. The proof relies on the monotonicity properties of $\mathbb{E}W_3(q, r)$, $\mathbb{E}W_4(q, r)$ and $\mathbb{E}W_5(q, r)$ which hold in the feasibility domain, i.e., if the effective arrival rates of queues 3, 4 and 5 are less than their respective service rates. This happens when $pq\lambda < \mu_3$, $(p(1-q) + r(1-p))\lambda < \mu_4$ and $(1-p)(1-r)\lambda < \mu_5$. We have already shown that each $(q, r) \in [0, 1]^2$ satisfies these three inequalities in the proof of Lemma A.1.

Therefore, it is true that $\mathbb{E}W_3(q, r)$, $\mathbb{E}W_4(q, r)$ and $\mathbb{E}W_5(q, r)$ are monotone on $[0, 1]^2$. In particular, $\mathbb{E}W_3(q, r)$ is increasing in q and constant in r , $\mathbb{E}W_4(q, r)$ is decreasing in q and increasing in r , $\mathbb{E}W_5(q, r)$ is constant in q and decreasing in r . These facts follow immediately from the fact that the waiting time in an M/M/1 queue is an increasing function of the arrival rate to the queue, as long the arrival rate is strictly less than the service rate.

It follows that f_1 is increasing in q and decreasing in r , and f_2 is decreasing in q and increasing in r . Now suppose (q_1, r_1) and (q_2, r_2) are two distinct zeros of f . We split the proof into two cases.

Case 1. $q_1 = q_2$

If this is the case, it must hold that $r_1 \neq r_2$. Then $0 = f_1(q_1, r_1) = f_1(q_2, r_1) \neq f_1(q_2, r_2) = 0$ as f_1 is decreasing in r , and we have a contradiction.

Case 2. without loss of generality, $q_2 > q_1$

We have for both i in $\{1, 2\}$ that $0 = f_1(q_i, r_i) = f_2(q_i, r_i)$. It follows that $0 = f_1(q_i, r_i) + f_2(q_i, r_i) = \mathbb{E}W_3(q_i, r_i) - \mathbb{E}W_5(q_i, r_i)$, so $\mathbb{E}W_3(q_i, r_i) = \mathbb{E}W_5(q_i, r_i)$, $i \in \{1, 2\}$. Since $\mathbb{E}W_3(q, r)$ is increasing in q and constant in r and $q_2 > q_1$, $\mathbb{E}W_3(q_2, r_2) = \mathbb{E}W_3(q_2, r_1) > \mathbb{E}W_3(q_1, r_1)$. Therefore, also $\mathbb{E}W_5(q_2, r_2) > \mathbb{E}W_5(q_1, r_1)$, and since $\mathbb{E}W_5(q, r)$ is constant in q , it holds that $\mathbb{E}W_5(q_2, r_1) = \mathbb{E}W_5(q_1, r_1) < \mathbb{E}W_5(q_2, r_2)$. So $\mathbb{E}W_5(q_2, r_1) < \mathbb{E}W_5(q_2, r_2)$, and as $\mathbb{E}W_5(q, r)$ is decreasing in r , it follows that $r_1 > r_2$.

As $f_1(q, r)$ is increasing in q and $q_2 > q_1$, $f_1(q_2, r_2) > f_1(q_1, r_2)$. As $f_1(q, r)$ is decreasing in r and $r_2 < r_1$, $f_1(q_1, r_2) > f_1(q_1, r_1)$. Therefore $f_1(q_2, r_2) > f_1(q_1, r_1)$, but this cannot be since $f_1(q_1, r_1) = f_1(q_2, r_2) = 0$. Therefore we have reached a contradiction.

To conclude, f must be injective in $[0, 1]^2$. $\square$

It now follows that F is injective on the domain $[-1, 1]^2$, and therefore the zero is unique.

We summarise these results in the following theorem. Note the two new conditions on the parameters in the theorem: they come from Lemma A.2. The conditions ensure that there is a change of sign in the function we apply the Poincaré-Miranda theorem to.

Theorem 4.9. If $p\lambda < \mu_3$, $\lambda(1-p) < \mu_5$, $\lambda < \mu_4$, $\frac{p\lambda}{\mu_3(\mu_3-p\lambda)} - \frac{(1-p)\lambda}{\mu_4(\mu_4-(1-p)\lambda)} > 0$ and $\frac{p\lambda}{\mu_4(\mu_4-p\lambda)} - \frac{(1-p)\lambda}{\mu_5(\mu_5-(1-p)\lambda)} < 0$, then a unique Nash equilibrium (p_N, q_N, r_N) exists for the non-zero routing problem.

These results provide us with conditions under which we can guarantee a unique Nash equilibrium. They are, however, not constructive.

4.1.2. The zero routing case

For certain combinations of parameters, it is possible that some servers, and therefore some routes, are not used by any customers in a Nash equilibrium. If this happens, we call the equilibrium a *zero routing equilibrium*. In this section, we will first show that if all service rates are finite, zero routing equilibria can only have two forms. In the first one, $q = 1$, and in the second $r = 0$. This is to say that $p, 1 - p, q$ and $1 - r$ cannot be equal to 0.

Lemma 4.10. *Suppose μ_i is finite for $i \in \{1, 2, 3, 4, 5\}$. If for a strategy s it holds that $p, 1 - p, q$ or $1 - r$ are 0, the strategy is not an equilibrium strategy.*

Proof. We have already seen in Lemma 4.2 that p cannot be 0 or 1 in equilibrium. Suppose now that we have a strategy s with $q = 0$. A tagged customer who has just finished service at server S_1 (such customers exist because $p \neq 0$) will have a waiting time of 0 in the second stage should she go to server S_3 , and a non-zero expected waiting time should she go to server S_4 . Therefore, her best response is again to deviate from the strategy, meaning that the strategy is not an equilibrium. Again, by an equivalent argument any strategy in which $r = 1$ is not a Nash equilibrium. $\square$

We therefore have two cases left. Suppose first that $q = 1$. This means that every customer that is done with service at server S_1 must choose to go to server S_3 rather than server S_4 . Therefore, $\mathbb{E}(W_3) < \mathbb{E}(W_4)$. The used routes are 1-3, 2-4, 2-5. Equating the waiting times on used routes, we get the following:

$$\begin{aligned}\mathbb{E}(W_1) + \mathbb{E}(W_3) &= \mathbb{E}(W_2) + \mathbb{E}(W_4), \\ \mathbb{E}(W_2) + \mathbb{E}(W_4) &= \mathbb{E}(W_2) + \mathbb{E}(W_5).\end{aligned}$$

The second equation gives $\mathbb{E}(W_4) = \mathbb{E}(W_5)$, and we have the following system of equations:

$$\frac{p\lambda}{\mu_1(\mu_1 - p\lambda)} + \frac{p\lambda}{\mu_3(\mu_3 - p\lambda)} = \frac{(1 - p)\lambda}{\mu_2(\mu_2 - (1 - p)\lambda)} + \frac{r(1 - p)\lambda}{\mu_4(\mu_4 - r(1 - p)\lambda)}, \quad (4.6)$$

$$\frac{r(1 - p)\lambda}{\mu_4(\mu_4 - r(1 - p)\lambda)} = \frac{(1 - p)(1 - r)\lambda}{\mu_5(\mu_5 - (1 - p)(1 - r)\lambda)}. \quad (4.7)$$

This system of equations could in theory be solved like the one in the previous subsection, but we would once again get a fourth degree polynomial in one of the unknowns. Even if we wrote out the analytical result, it would be too cumbersome to gain any insight from it. Therefore, we make a numerical solver which solves the system using an iterative scheme instead.

We do the same for the equilibrium in which $r = 0$. In this case, a customer that is done with service at server S_2 must always choose going into queue 5 rather than queue 4 - it must hold that $\mathbb{E}(W_5) < \mathbb{E}(W_4)$. In that case, the used routes are 1-3, 1-4, 2-5. The equations are

$$\begin{aligned}\mathbb{E}(W_3) &= \mathbb{E}(W_4), \\ \mathbb{E}(W_1) + \mathbb{E}(W_4) &= \mathbb{E}(W_2) + \mathbb{E}(W_5),\end{aligned}$$

or once we plug in the formulas for the expected waiting times,

$$\frac{pq\lambda}{\mu_3(\mu_3 - pq\lambda)} = \frac{p(1-q)\lambda}{\mu_4(\mu_4 - p(1-q)\lambda)}, \quad (4.8)$$

$$\frac{p\lambda}{\mu_1(\mu_1 - p\lambda)} + \frac{p(1-q)\lambda}{\mu_4(\mu_4 - p(1-q)\lambda)} = \frac{(1-p)\lambda}{\mu_2(\mu_2 - (1-p)\lambda)} + \frac{(1-p)\lambda}{\mu_5(\mu_5 - (1-p)\lambda)}. \quad (4.9)$$

These equations will be solved numerically.

Next we prove a simple result which will be useful when numerically solving for the Nash equilibrium. Given parameters of the network and an initial guess for the routing probabilities, we aim to produce the Nash routing probabilities. But before doing so, it must be determined to which of the three cases the solution belongs: the zero routing case with $q = 1$, the zero routing case with $r = 0$, or the non-zero routing case. This is done using the following lemma.

Lemma 4.11. *Suppose that we have Nash equilibrium routing probabilities (p, q, r) . If $\mathbb{E}(W_1) = \mathbb{E}(W_2)$, then it is a non-zero routing equilibrium. If $\mathbb{E}(W_1) < \mathbb{E}(W_2)$, then it is a zero routing equilibrium with $r = 0$. If $\mathbb{E}(W_1) > \mathbb{E}(W_2)$, then it is a zero routing equilibrium with $q = 1$.*

Proof. First, note that in each of the scenarios, the routes 1 – 3 and 2 – 5 are used, and therefore the waiting times of these routes must be equal. In an equation, this is

$$\mathbb{E}(W_1) + \mathbb{E}(W_3) = \mathbb{E}(W_2) + \mathbb{E}(W_5). \quad (4.10)$$

If $q = 1$, then $\mathbb{E}(W_4) = \mathbb{E}(W_5)$ and $\mathbb{E}(W_4) > \mathbb{E}(W_3)$, since otherwise q would not be equal to 1. Therefore, $\mathbb{E}(W_5) > \mathbb{E}(W_3)$. Similarly, if $r = 0$, $\mathbb{E}(W_3) = \mathbb{E}(W_4)$ and $\mathbb{E}(W_4) > \mathbb{E}(W_5)$, since otherwise r would not be 0. Finally, if neither $r = 0$ nor $q = 1$, it holds that $\mathbb{E}(W_3) = \mathbb{E}(W_4) = \mathbb{E}(W_5)$.

Suppose $\mathbb{E}(W_1) = \mathbb{E}(W_2)$. Then from Equation (4.10), we see that $\mathbb{E}(W_3) = \mathbb{E}(W_5)$, so we cannot be in either of the zero routing cases and are therefore in the non-zero routing case. If $\mathbb{E}(W_1) > \mathbb{E}(W_2)$, Equation (4.10) implies $\mathbb{E}(W_5) > \mathbb{E}(W_3)$, which then implies that $q = 1$. Similarly, if $\mathbb{E}(W_1) < \mathbb{E}(W_2)$, Equation (4.10) implies $\mathbb{E}(W_5) < \mathbb{E}(W_3)$, which then implies that $r = 0$. $\square$

Given an initial guess and the parameters, we solve for each case the system of equations that characterises it. For the non-zero routing case, these are Equations (4.1), (4.2) and (4.3). We minimise the difference between the left hand side and the right hand side

of the equations. The result of solving this minimisation problem is a triple (p_1, q_1, r_1) . For the zero routing case with $q = 1$, the equations we want to solve are Equations (4.6) and (4.7). This is again phrased as a minimisation problem, and probabilities (p_2, q_2, r_2) are obtained, where $q_2 = 1$. Finally, for the zero routing case with $r = 0$, we solve Equations (4.8) and (4.9) and obtain a triple (p_3, q_3, r_3) with $r_3 = 0$. For all three triples, the quantity $\mathbb{E}(W_1) - \mathbb{E}(W_2)$ is computed, and a check is made to see whether this quantity has the sign it should according to the above lemma. If only one triple passes the check, it is returned as the solution. If more than one triple passes the check, the minimisation problems are solved again, but with a slightly smaller tolerance. If no triples pass the check, the minimisation problems are solved again, with a slightly larger tolerance.

4.2. Social Optimum

Aside from the individually optimal strategies, we are also interested in those strategies that minimise the social cost, which in the unobservable case, we define as

$$SC(p, q, r) = \sum_{R_i} \lambda_{R_i} \mathbb{E}(W_{R_i}),$$

where R_i is a route in the network, and λ_{R_i} is the arrival rate to route R_i . Hence, the social cost is the sum of the waiting times on each route, weighted by the number of customers joining this route. Intuitively, a social planner's job will then be to perform a sort of load balancing on the routes, so that the average customer has to wait as little as possible.

The expression for the social cost further rewrites to

$$\begin{aligned} SC(p, q, r) &= \lambda pq [\mathbb{E}(W_1) + \mathbb{E}(W_3)] + \lambda p(1 - q) [\mathbb{E}(W_1) + \mathbb{E}(W_4)] \\ &\quad + \lambda(1 - p)r [\mathbb{E}(W_2) + \mathbb{E}(W_4)] + \lambda(1 - p)(1 - r) [\mathbb{E}(W_2) + \mathbb{E}(W_5)] \\ &= \lambda p \mathbb{E}(W_1) + \lambda(1 - p) \mathbb{E}(W_2) + \lambda pq \mathbb{E}(W_3) \\ &\quad + \lambda((1 - p)r + (1 - q)p) \mathbb{E}(W_4) + \lambda(1 - r)(1 - p) \mathbb{E}(W_5). \end{aligned}$$

Plugging in the expressions for the expected waiting times at each queue, we arrive at the following expression for the social cost

$$\begin{aligned} SC(p, q, r) &= \frac{(\lambda p)^2}{\mu_1(\mu_1 - \lambda p)} + \frac{(\lambda(1 - p))^2}{\mu_2(\mu_2 - \lambda(1 - p))} + \frac{(\lambda pq)^2}{\mu_3(\mu_3 - \lambda pq)} \\ &\quad + \frac{(\lambda(p(1 - q) + (1 - p)r))^2}{\mu_4(\mu_4 - \lambda(p(1 - q) + (1 - p)r))} + \frac{(\lambda(1 - r)(1 - p))^2}{\mu_5(\mu_5 - \lambda(1 - r)(1 - p))}. \end{aligned}$$

The goal is to find a (p, q, r) that minimises the social cost, subject to the feasibility constraints. Formally, we have the following optimisation problem

$$\begin{aligned}
& \min_{(p,q,r)^T} \quad \frac{(\lambda p)^2}{\mu_1(\mu_1 - \lambda p)} + \frac{(\lambda(1-p))^2}{\mu_2(\mu_2 - \lambda(1-p))} + \frac{(\lambda pq)^2}{\mu_3(\mu_3 - \lambda pq)} \\
& \quad + \frac{(\lambda(p(1-q) + (1-p)r))^2}{\mu_4(\mu_4 - \lambda(p(1-q) + (1-p)r))} + \frac{(\lambda(1-r)(1-p))^2}{\mu_5(\mu_5 - \lambda(1-r)(1-p))} \\
& \text{s.t.} \quad \lambda p < \mu_1 \quad (4.11) \\
& \quad \lambda(1-p) < \mu_2 \quad (4.12) \\
& \quad \lambda pq < \mu_3 \quad (4.13) \\
& \quad \lambda(p(1-q) + r(1-p)) < \mu_4 \quad (4.14) \\
& \quad \lambda(1-p)(1-r) < \mu_5 \quad (4.15) \\
& \quad (p, q, r)^T \in [0, 1]^3 \quad (4.16)
\end{aligned}$$

Taking the partial derivatives of the social cost with respect to p, q and r and setting them to zero, we once again obtain a nonlinear system of three equations. We could solve these three equations and check that they satisfy the feasibility criteria given by Inequalities (4.11), (4.12), (4.13), (4.14), (4.15) and (4.16). However, in contrast with the equations we obtained for Nash, all three equations involve all three variables. Because of this, we solve the optimisation problem numerically.

In the next section, we present results from the numerical solvers. These include both the Nash and the social optimum solver.

5. Unobservable Case - Numerical Observations

In this chapter we study the Nash and socially optimal strategy numerically. In Section 5.1, we study the influence of the service rates on the strategies. We show when zero routing occurs, and how the price of anarchy depends on the service rates. In Section 5.2 we study the influence of the arrival rate on the strategies. Finally, we show an interesting result regarding bottlenecks in Section 5.3.

5.1. Influence of service rates on the Nash strategy

We begin the numerical analysis by investigating how varying one service rates impacts the strategy, when the other service rates are fixed. We vary first μ_1 , then μ_3 , and finally μ_4 . The impact of μ_2 and μ_5 follows by symmetry. Areas where zero routing occurs are shaded grey in the plots.

Figure 5.1 shows the socially optimal and Nash routing probabilities against μ_1 which varies from 0 to 6. The service rate at S_2 is equal to 1.5, the arrival rate is equal to 1.4, and the other service rates are equal to 1. The most striking impression we get is that the socially optimal and Nash routing probabilities are very close to each other. At least in this case, selfish customers follow a near-optimal strategy. Unsurprisingly, as μ_1 grows, so does p , the probability of opting for server S_1 upon arrival. When μ_1 is small, zero routing with q equal to 1 occurs. This is clear, as in that case there will be very few customers using server S_1 compared to ones using server S_2 , and it will therefore be profitable for all server S_1 customers to go to server S_3 , since its service is just as fast as server S_4 , but its arrival process is slower. On the other side of this spectrum, if μ_1 is very large, r goes to 0. This is an equivalent phenomenon: now the rate of server S_2 is comparatively slow, so customers done with service at S_2 will all go to S_5 . When $q = 1$, r is exactly equal to 0.5. This also makes sense, since in that case customers that done with service at server S_2 have two servers in front of them which have equal service rate, and are only used by these customers. Thus, there is equal splitting of customers between these two servers. The same conclusion holds for zero routing with $r = 0$, there q is equal to 0.5.

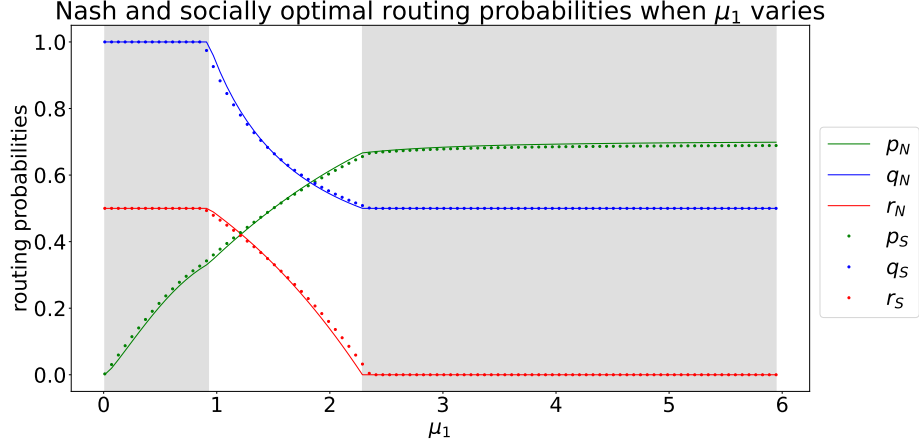


Figure 5.1: Nash and socially optimal routing probabilities p, q , and r as a function of parameter $\mu = \mu_1$. Zero routing areas are shaded grey. In this figure, $\lambda = 1.4, \mu_2 = 1.5, \mu_3 = \mu_4 = \mu_5 = 1$.

Figure 5.2 shows the same routing probabilities, only now μ_3 varies. The arrival rate is 1, the first stage service rates are equal to 1.5 and the other second stage service rates are equal to 1. Again, the selfish routing probabilities are near optimal. In the non-zero routing area, $\mu_1 = \mu_2$ and thus $p = 0.5$. As μ_3 grows, q grows as well, and finally settles at the value 1 when we reach the zero routing region. Again, r is equal to 0.5 here, and p grows with μ_3 , as we expect.

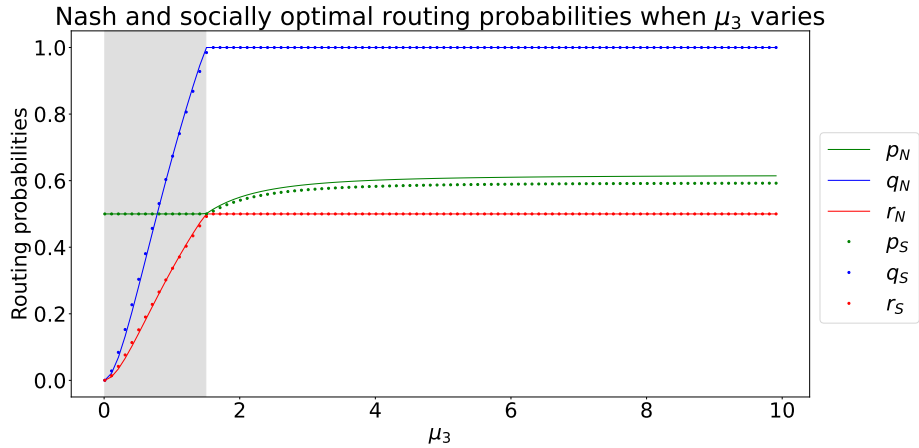


Figure 5.2: Nash and socially optimal routing probabilities p, q , and r as a function of parameter $\mu = \mu_3$. Zero routing areas are shaded grey. In this figure, $\lambda = 1, \mu_1 = \mu_2 = 1.5, \mu_4 = \mu_5 = 1$.

In Figure 5.3, the arrival rate is equal to 1.4, the first stage service rates are equal to 1.5, $\mu_3 = \mu_5 = 1$. We vary μ_4 and plot the routing probabilities. There is no zero

routing. This is in line with our expectations: as μ_4 increases, the waiting time at S_4 will be lower. In order to maintain equality with the other waiting times, the arrival rate to this server will increase, i.e., r and $1 - q$ will increase. As the routing probabilities $1 - q$ and r can never be 1 in an equilibrium, we expect continuous curves with diminishing returns, just as the figure shows.

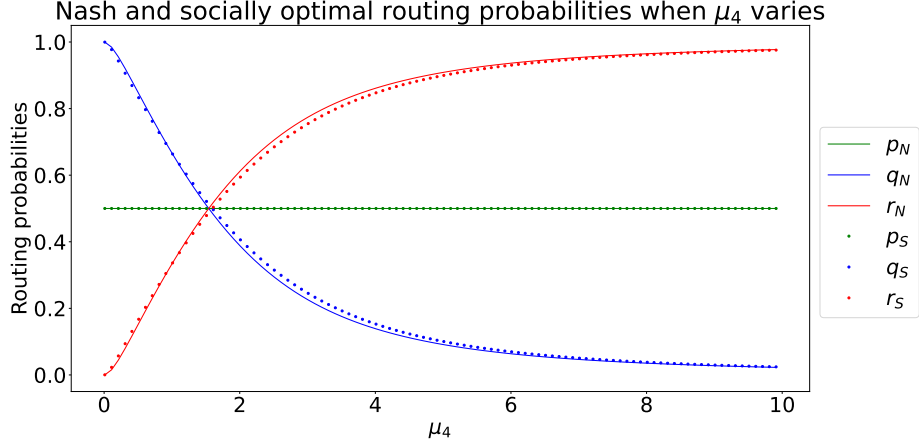


Figure 5.3: Nash and socially optimal routing probabilities p, q , and r as a function of parameter $\mu = \mu_4$. Zero routing areas are shaded grey. In this figure, $\lambda = 1.4, \mu_1 = \mu_2 = 1.5, \mu_3 = \mu_5 = 1$.

We can already see from these plots that the Nash and socially optimal routing probabilities are close. We investigate this further in Figures 5.4, 5.5, and 5.6, where instead of plotting the routing probabilities, we plot the difference between the social and Nash probabilities. The parameters are as before, so for example in Figure 5.4, $\lambda = 1.4, \mu_2 = 1.5, \mu_3 = \mu_4 = \mu_5 = 1$.

Figure 5.4 shows that this difference as we vary μ_1 . The difference is precisely 0 when $\mu_1 = \mu_2$. The intuition is that in this case, the system is perfectly balanced (as also $\mu_3 = \mu_4 = \mu_5$) and there is no incentive to deviate from the social optimum.

Figure 5.5, in which we vary μ_3 , shows that when μ_3 becomes very large and we enter the zero routing area, $q_N = q_S$ and $r_N = r_S$. It seems that at one point the social planner will also want to route all customers from server S_1 to server S_3 .

In all three plots, the differences between the socially optimal and the Nash routing probabilities are small. Moreover, in all three plots we see that the differences tend to 0 whenever the varying parameter tends to 0. This effectively means removing the server from the network, which in some cases removes some choices, implying that the Nash and social strategy are equal. For example, when $\mu_4 = 0$, the only choice that remains is whether to join S_1 or S_2 upon arrival. In the case of Figure 5.6, we have set $\mu_1 = \mu_2$ and $\mu_3 = \mu_5$, and the network is symmetric, so it should not be surprising that both in social optimum and Nash, $p = 0.5$.

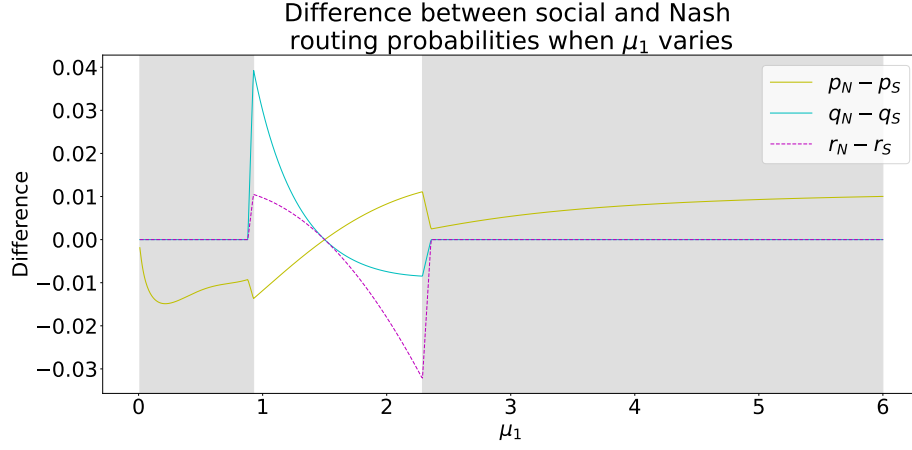


Figure 5.4: Difference between Nash and socially optimal routing probabilities p , q , and r as a function of parameter $\mu = \mu_1$. Zero routing areas are shaded grey. In this figure, $\lambda = 1.4$, $\mu_2 = 1.5$, $\mu_3 = \mu_4 = \mu_5 = 1$.

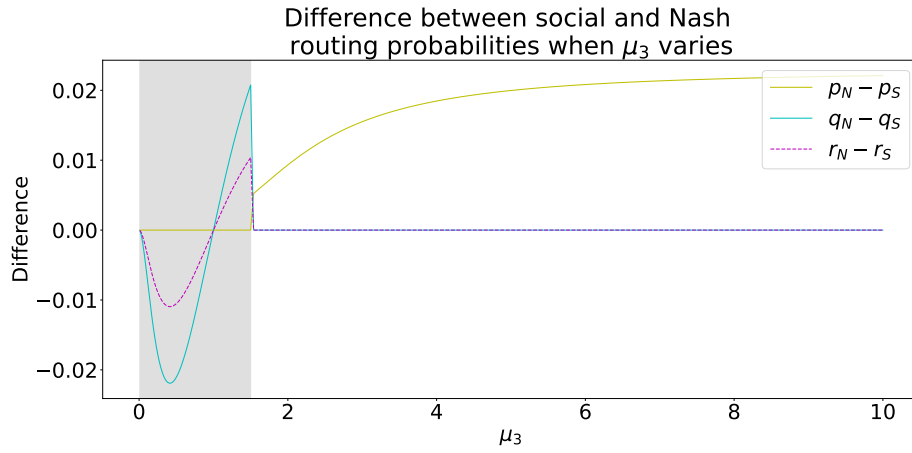


Figure 5.5: Difference between Nash and socially optimal routing probabilities p , q , and r as a function of parameter $\mu = \mu_3$. Zero routing areas are shaded grey. In this figure, $\lambda = 1$, $\mu_1 = \mu_2 = 1.5$, $\mu_4 = \mu_5 = 1$.

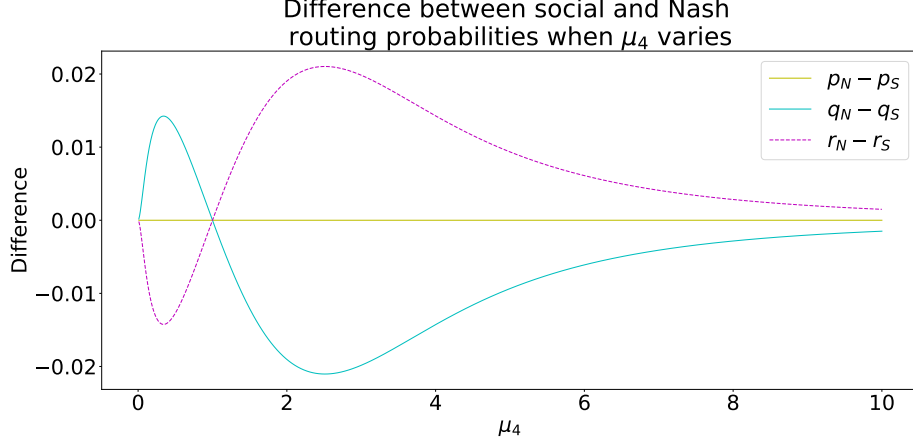


Figure 5.6: Difference between Nash and socially optimal routing probabilities p , q , and r as a function of parameter $\mu = \mu_4$. Zero routing areas are shaded grey. In this figure, $\lambda = 1.4$, $\mu_1 = \mu_2 = 1.5$, $\mu_3 = \mu_5 = 1$.

We also make graphs of the price of anarchy as a function of the varying parameter, shown in Figures 5.7, 5.8, and 5.9. The PoA attains its minimal value either when the varying service rate is 0, or when the service rates at each stage of the network are equal. This is clear since at these points the routing probabilities coincide in Nash and the social optimum. Figures 5.7 and 5.8 show that the PoA has two maxima, one between values 0 and 1, and another one, attained at infinity. The second maximum is in line with what we saw previously, as in these cases $r_N = r_S$, $q_N = q_S$, and the difference between p_N and p_S grows steadily. Figure 5.9 shows two maxima, one around $\mu_4 = 0.5$ and one around $\mu_4 = 2$. In all three plots, we see that the PoA is quite small, meaning that using social planning does not have a lot of added value.

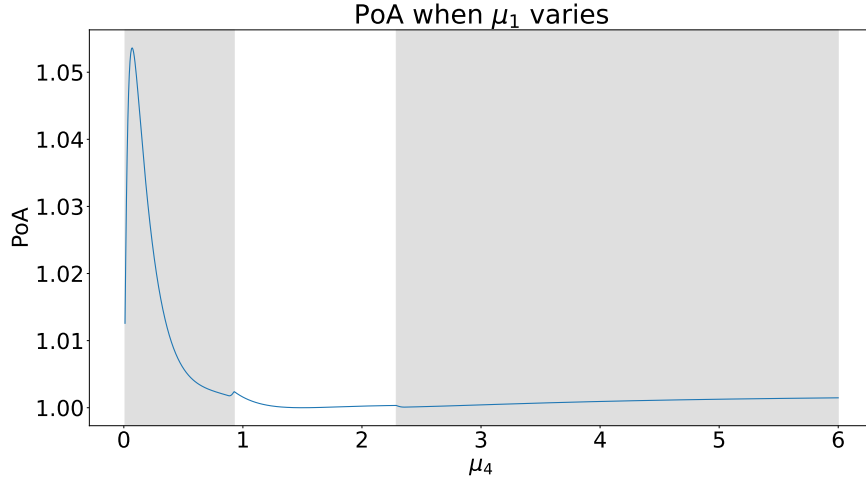


Figure 5.7: Price of anarchy as a function of parameter $\mu = \mu_1$. Zero routing areas are shaded grey. In this figure, $\lambda = 1.4, \mu_2 = 1.5, \mu_3 = \mu_4 = \mu_5 = 1$.

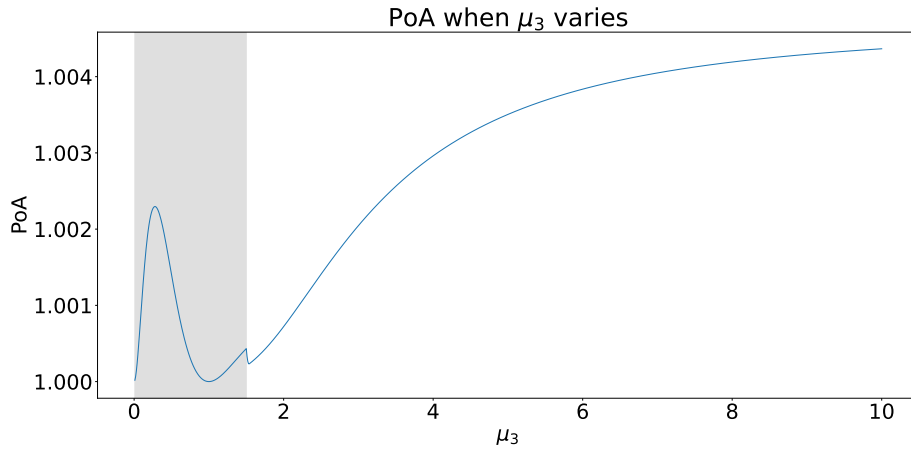


Figure 5.8: Price of anarchy as a function of parameter $\mu = \mu_3$. Zero routing areas are shaded grey. In this figure, $\lambda = 1, \mu_1 = \mu_2 = 1.5, \mu_4 = \mu_5 = 1$.

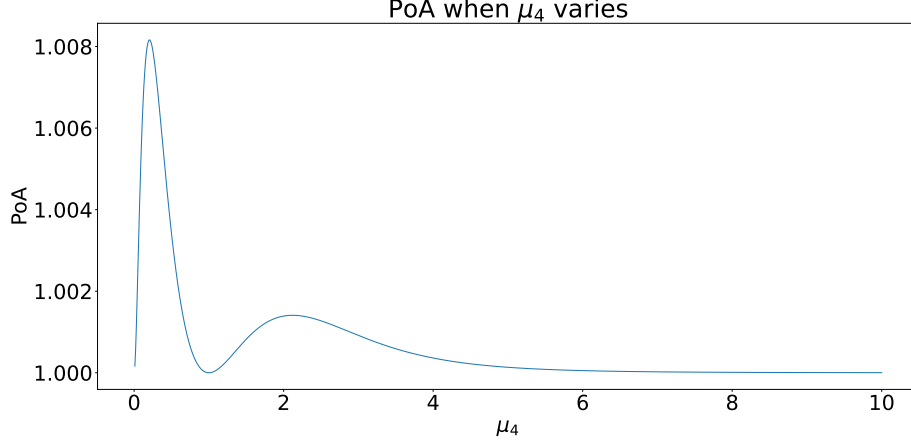


Figure 5.9: Price of anarchy as a function of parameter $\mu = \mu_4$. Zero routing areas are shaded grey. In this figure, $\lambda = 1.4, \mu_1 = \mu_2 = 1.5, \mu_3 = \mu_5 = 1$.

We have now seen that the PoA is the smallest when the service rates are equal within each stage. Perhaps of more interest is the social cost, since it is a measure of how well the network is performing. We have plotted the social cost in Nash and social optimum in Figure 5.10. Unsurprisingly, it decreases with increasing server speed.

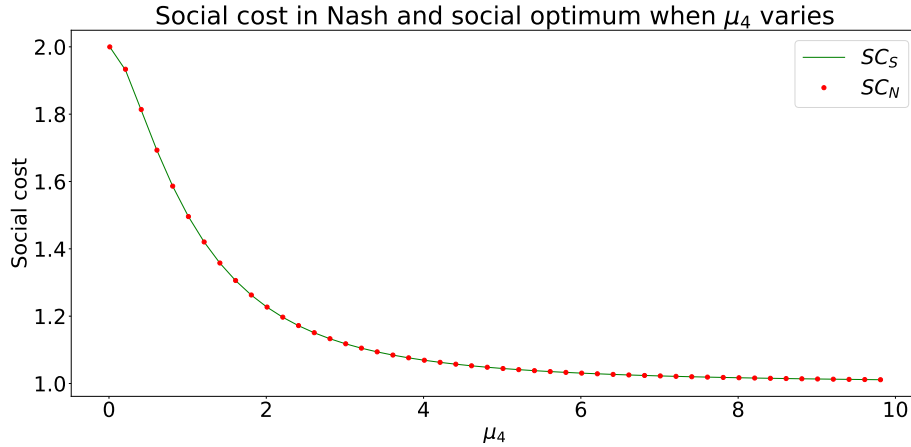


Figure 5.10: Social cost is a decreasing function of μ_4 . In this figure, $\lambda = 1, \mu_1 = \mu_2 = 1.5, \mu_3 = \mu_5 = 1$.

In order to confirm our suspicion that the PoA is minimised when the servers within each stage operate at an equal rate, we did the following. We took 20 equally spaced points in $[1, 6]$. For each of these values, we set μ_2 to this value and we set $\mu_3 = \mu_4 = \mu_5$ to a randomly sampled number between 0 and 10. We then find the corresponding μ_1 that minimises the PoA. The results are shown in Figure 5.11. The plot shows that the PoA is minimised precisely when $\mu_1 = \mu_2$, as we suspected. This is only the case when

$\mu_3 = \mu_4 = \mu_5$, otherwise the system dynamics are different and we can make no such claim.

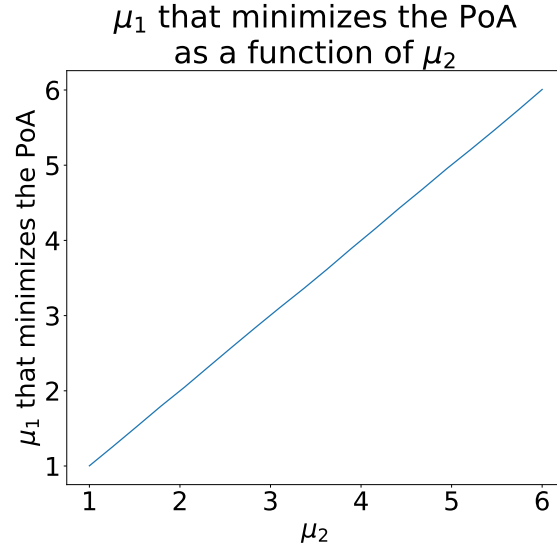


Figure 5.11: The PoA is minimised when service rates are equal within each stage. In this figure, $\lambda = 0.9$, $\mu_3 = \mu_3 = \mu_4$ randomly sampled.

5.2. Influence of the arrival rate on the Nash strategy

We also investigate how varying the arrival rate impacts the results. We set $\mu_1 = \mu_2 = \mu_3 = \mu_5 = 1$ and $\mu_4 = 2$, and vary λ between 0 and 2. Figure 5.12 shows the Nash and socially optimal routing probabilities.

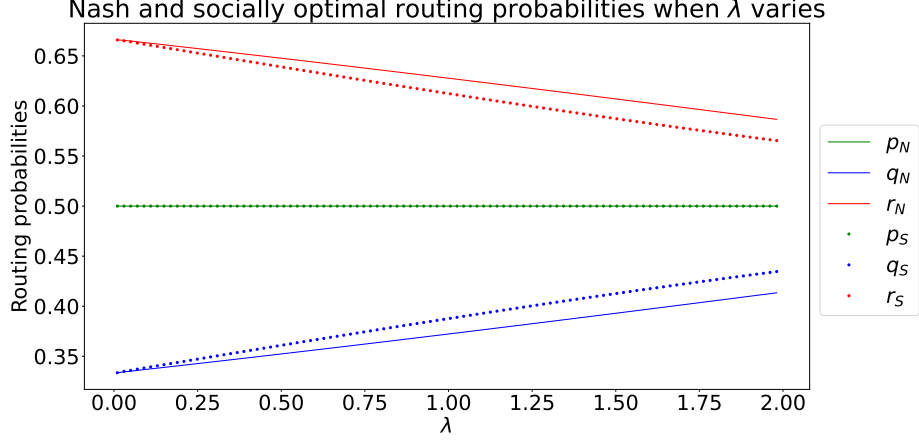


Figure 5.12: Nash routing probabilities as a function of λ . In this figure, $\mu_1 = \mu_2 = \mu_3 = \mu_5 = 1, \mu_4 = 2$.

As $\mu_1 = \mu_2$ and all routes are used, Lemma 4.5 gives $p_N = 0.5$. It also makes sense intuitively that $p_S = 0.5$, since social optimisation is a form of load balancing, and there is a lot of symmetry in the network, namely $\mu_1 = \mu_2$ and $\mu_3 = \mu_4$. We also see that $r_S < r_N, q_S > q_N$, meaning that in the social optimum more customers are directed to the slower servers. Our intuition here is that selfish customers choose to go to the server with a higher service rate more aggressively, compared to the social customers. It would also make sense that this effect is more pronounced when λ increases, as the waiting times grow larger overall and the faster servers become even more appealing to selfish customers.

Figure 5.14 shows the social cost in Nash and social optimum for $0.01 < \lambda < 1.95$. The social cost tends to infinity as λ increases to 2, as we would expect. When λ is equal to $2 - 10^3$, both Nash and the social optimum give a value of around 4000. This can perhaps explain why Figure 5.13 shows that the PoA goes to 1 as λ goes to 2. Neither strategy can stabilise the system any more, and both social costs grow very quickly.

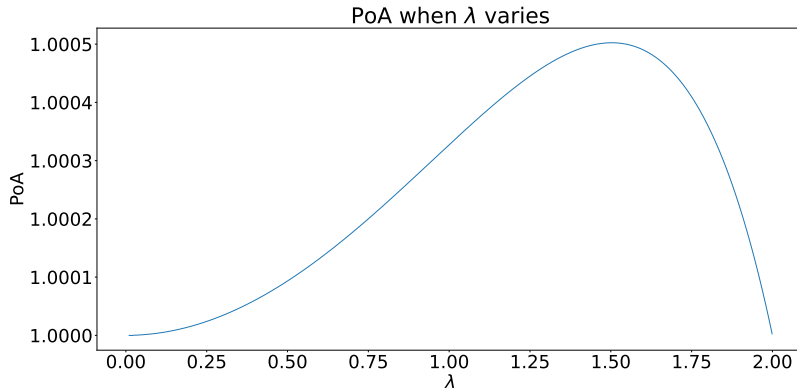


Figure 5.13: PoA as a function of λ . In this figure, $\mu_1 = \mu_2 = \mu_3 = \mu_5 = 1, \mu_4 = 2$.

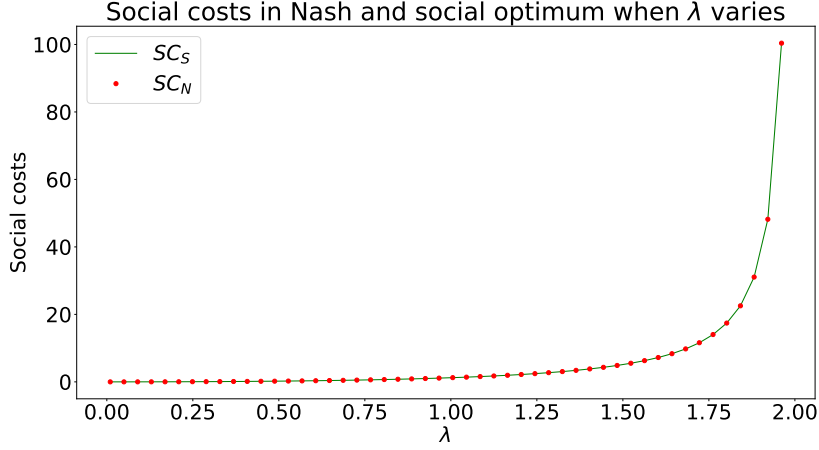


Figure 5.14: The social cost in Nash and social optimum as a function of λ . In this figure, $\mu_1 = \mu_2 = \mu_3 = \mu_5 = 1, \mu_4 = 2$.

5.3. Social cost in a bottleneck

In this section we present a nice managerial insight regarding a bottleneck situation, where the first stage service rates are smaller than the second stage service rates. We observe that if this is the case, it is best (in terms of reducing social cost) to increase the service rate of the second slowest server. More formally, we made a loop of length 100.

In each iteration, we first set μ_1 and μ_2 equal to two randomly sampled values from the interval $[0, 1]$ and all the stage two service rates equal to one randomly sampled value from the interval $[3, 4]$. This way there is a bottleneck in the first stage. We set $\lambda = 0.9 \min\{\mu_1 + \mu_2, \mu_3 + \mu_4 + \mu_5\}$, so that the system load is quite high, as we are most interested in bottlenecks when there is also congestion in the system. Next, we compute the social cost in the Nash equilibrium. We call this the original social cost. We then run 5 experiments. In each experiment, we increase one of the five service rates by 1, and record the new Nash social cost. We record in which of the five experiments, the difference between the original social cost and the decreased social cost is most prominent. In each of the 100 iterations, the answer is always the same: the Nash social cost is reduced most when we increase the second slowest server in the first stage. This is to say, rather than trying to resolve the bottleneck, it is seemingly more profitable to increase the speed of the faster server in the bottleneck. We then repeat the experiment, but consider the social cost in the social optimum rather than in the Nash equilibrium. The same result is observed.

6. Fully Observable Case - Analytic Solutions

The topic of this chapter is the observable case, where joining customers know the network parameters and how long the queue at each station is. We begin by describing the problem and the difficulties that arise in this setting, compared to the unobservable case. After formalising the model, we analyse in Section 6.1 a simplified version of the problem which is obtained by removing server S_4 . We provide an exact solution to the simplified problem along with some approximations. Next, in Section 6.2 we analyse the full problem. We do this by performing policy iteration. At each step of the iteration, we write the problem as a high-dimensional linear system of equations and find an approximate solution using a numerical scheme.

Going back to the problem at hand, the network is shown in Figure 6.1. Upon arrival to the system, a customer observes state $(n_1, n_2, n_3, n_4, n_5, 0)$, where n_i is the number of customers at station i , and 0 denotes that the customer has just arrived to the system. The customer decides which of the two queues to join by minimising her expected waiting time in the system. After receiving service at her first stage server of choice, she again observes how many customers are at each station, and the new state she sees is $(n'_1, n'_2, n'_3, n'_4, n'_5, i)$. Here $i = 1$ if the customer just obtained service at S_1 , and $i = 2$ if she just obtained service at S_2 . Once again the customer chooses which queue to join by minimising her expected waiting time. The state space is now $\mathbb{Z}_{\geq 0}^5 \times \{0, 1, 2\}$. The action spaces depend only on the customer's position in the network, hence we denote them as $\mathcal{A}_0 = \{1, 2\}$, $\mathcal{A}_1 = \{3, 4\}$, and $\mathcal{A}_2 = \{4, 5\}$. A strategy s maps each state $(n_1, n_2, n_3, n_4, n_5, i)$ to $\mathcal{A}_i$. A Nash equilibrium strategy is a strategy s_N such that if everyone uses this strategy, then the expected waiting time of each customer is precisely minimised when she also uses this strategy.

It is not difficult to see what this strategy prescribes once a customer has finished service at one of the first stage servers. Let us suppose that the customer went to server S_1 , and the state she observes upon service completion is $(n_1, n_2, n_3, n_4, n_5, 1)$. Then her expected remaining waiting time in the system is n_3/μ_3 if she joins queue 3, or n_4/μ_4 if she joins queue 4. Therefore, the Nash strategy in this state is to join queue 3 if $\frac{n_3}{\mu_3} < \frac{n_4}{\mu_4}$ and join queue 4 if $\frac{n_3}{\mu_3} > \frac{n_4}{\mu_4}$. If the expected waiting times at both queues are precisely equal, the customer can join queue 3 with some probability $p \in [0, 1]$, and join queue 4 with probability $1 - p$. Any choice of $p \in [0, 1]$ results in minimal expected waiting time. We will call this strategy the *weighted join shortest queue* strategy with parameter p , abbreviated WJSQ(p). It prescribes an individually optimal decision to a problem with two observable queues in parallel.

Now that we have described what the Nash equilibrium strategy is for the second stage decisions, let us explain why the first stage decision is difficult. Suppose a customer has just arrived to a system, she observed state $(n_1, n_2, n_3, n_4, n_5)$, and she must choose whether to join queue 1 or queue 2. To do this, she must compute $\mathbb{E}(W|(n_1, n_2, n_3, n_4, n_5), S_1)$ and $\mathbb{E}(W|(n_1, n_2, n_3, n_4, n_5), S_2)$, her expected waiting times. If she goes to server S_1 , she knows she will have to wait (in expectation) n_1/μ_1 time units in the first stage. Her second stage expected waiting time is simple if she knows what state she will see once she is done with service at S_1 , but this prediction is quite difficult to make. We would need to know where each customer that is in front of her in queue 1 went upon service completion at S_1 . To know this, we need to predict the state of the network at each service completion.

To make matters worse, there is also the possibility of overtaking: while the tagged customer is waiting for service in queue 1, a new arrival may come into the network, go to queue 2, and then join queue 4 before the tagged customer. Therefore, a tagged customer's decision depends on future customers, whose decision in turn depends on the tagged customer's decision. Another problem is that the state space is unbounded. This means that when we want to solve the problem numerically, we will have to truncate the state space, which will result in an approximation. The state space is also five-dimensional, which means that the so-called curse of dimensionality will be an issue.

Because of the challenges outlined above, we first studied a simplified network, obtained from the full network by removing S_4 . In doing so, we remove the possibility of overtaking as well as the second stage decision. The simplified network is the topic of the following section.

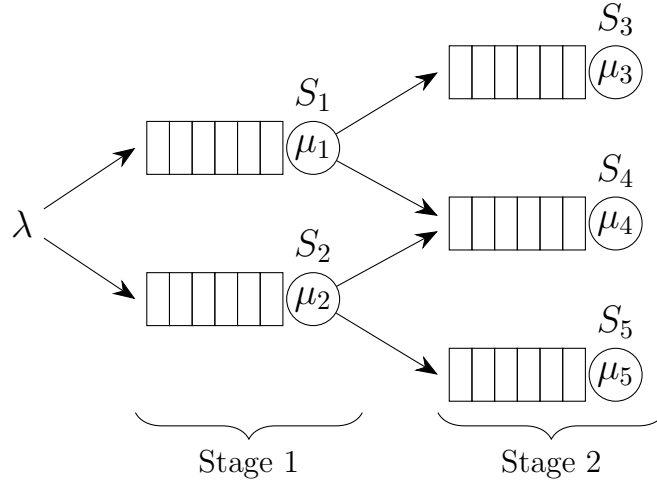


Figure 6.1: The network in the observable case.

6.1. The Simplified Model

The network, after removing S_4 , is shown in Figure 6.2. Here, S_5 has been relabelled to S_4 for convenience. Removing the middle second stage server also removes some of the

issues we discussed at the beginning of this chapter.

Firstly, there is only one decision now, and this is which of the two tandems to join. The problem is then to determine which queue to join when the arriving customer observes queue length $x_0 = (n_1, n_2, n_3, n_4)$ upon arrival. She joins S_1 if $\mathbb{E}(W|x_0, S_1) < \mathbb{E}(W|x_0, S_2)$ and she joins S_2 if $\mathbb{E}(W|x_0, S_2) < \mathbb{E}(W|x_0, S_1)$. If the two expected waiting times are equal, the customer joins queue 1 with probability $p \in [0, 1]$, and queue 2 with probability $1 - p$.

The second advantage of removing the middle second stage server is that there is no possibility of overtaking in the resulting network. This fact has two implications. The first implication is that once a customer joins one of the first stage queues, the progress of the other tandem no longer influences her expected waiting times. Therefore, we can reduce the needed expressions $\mathbb{E}(W|x_0, S_1)$ and $\mathbb{E}(W|x_0, S_2)$ to $\mathbb{E}(W|(n_1, n_3), S_1)$ and $\mathbb{E}(W|(n_2, n_4), S_2)$, respectively. The second implication is that future arrivals no longer contribute to the expected waiting times, and therefore also do not influence the equilibrium strategy. This means that we can assume there are no arrivals after the tagged customer, and the equilibrium strategy will not change - a simplification that will be useful when we want to describe the evolution of the network.

Finally, the removal of the second stage decision means that predicting the state of the network in some point in time becomes much easier. This will be of particular use in the so-called “no starvation” approximation, which is the topic of the following subsection.

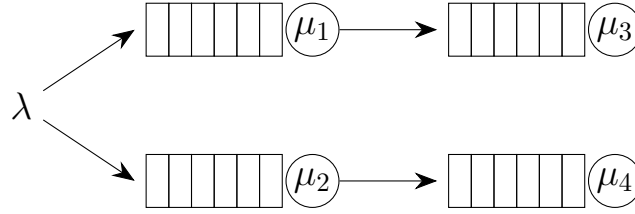


Figure 6.2: The simplified network.

6.1.1. The “No Starvation” Approximation of the Nash Equilibrium Strategy

Suppose a customer arrives to the network and observes state (n_1, n_2, n_3, n_4) . Suppose also that she joins queue 1. Let T be the random variable that is her sojourn time in the first stage of the network. Let L be the random variable that is the number of customers she sees in front of her at station 3, once she has obtained service at S_1 . Note that we suppress the dependence of L on n_1 and n_3 . The discrete random variable L takes values in the set $\{0, 1, \dots, n_3 + n_1\}$; we denote its probability mass function by p_L . Then we may condition on L the total expected waiting time of the tagged customer as follows

$$\mathbb{E}(W|(n_1, n_2, n_3, n_4), S_1) = \frac{n_1}{\mu_1} + \sum_{l=0}^{n_1+n_3} \frac{l}{\mu_3} p_L(l).$$

Let L_t be the event that the tagged customer's sojourn time in queue 1 is t and she sees l customers in queue 3 upon her service completion at S_1 . Then $p_L(l) = \mathbb{P}(L = l) = \mathbb{P}(\cup_{t \in \mathbb{R}_{\geq 0}} L_t)$. As the events L_t are mutually exclusive, we can write $\mathbb{P}(\cup_{t \in \mathbb{R}_{\geq 0}} L_t) = \int_0^\infty \mathbb{P}(L_t) dt$.

The event L_t is an intersection of two events: $A_t = \{\text{tagged customer's sojourn time in queue 1 takes } t \text{ time}\}$, and $B = \{n_1 + n_3 - l \text{ services occur at } S_3 \text{ during the tagged customer's sojourn time in queue 1}\}$. Then $B|A_t = \{n_1 + n_3 - l \text{ services occur at } S_3 \text{ during time } t\}$.

We can then write $\mathbb{P}(L_t) = \mathbb{P}(B|A_t)\mathbb{P}(A_t)$, and rewrite the density p_L as

$$p_L(l) = \mathbb{P}(\cup_{t \in \mathbb{R}_{\geq 0}} L_t) = \int_0^\infty \mathbb{P}(B|A_t)\mathbb{P}(A_t)dt. \quad (6.1)$$

The probability that the tagged customer's sojourn time in queue 1 is equal to some value t is simply the probability that $n_1 + 1$ services at S_1 take exactly t time units. The service durations are exponential random variables with rate μ_1 , so the sum of $n_1 + 1$ service times has an Erlang distribution with shape parameter $n_1 + 1$ and rate parameter μ_1 . It remains to determine the probability that $n_1 + n_3 - l$ services occur at S_3 during time t . This situation is reminiscent of a Poisson distribution: there is a fixed time interval of length t and we count the number of events that occur within this interval. Note, however, that the rate at which the events occur is not fixed. Once the initial n_3 customers at S_3 have been served, it may happen that the server S_3 is starved. This happens when there are no arrivals to S_3 , because S_1 is still processing the customers. This approximation ignores this possibility and assumes that S_3 is never starved.

Once this simplification is made, events occur at rate μ_3 , and the probability of $B|A_t$ is simply the probability mass function of a $\text{Poisson}(\mu_3 t)$ random variable, evaluated at $n_1 + n_3 - l$. Plugging in the Poisson probability mass function and the Erlang probability density function into Equation (6.1), the following is obtained

$$\begin{aligned} \hat{p}_L(l) &= \int_0^\infty \left(\frac{\mu_1^{n_1+1} t^{n_1}}{n_1!} e^{-\mu_1 t} \right) \left(\frac{(\mu_3 t)^{n_1+n_3-l}}{(n_1+n_3-l)!} e^{-\mu_3 t} \right) dt \\ &= \frac{\mu_1^{n_1+1} \mu_3^{n_1+n_3-l}}{n_1!(n_1+n_3-l)!} \int_0^\infty t^{2n_1+n_3-l} e^{-(\mu_1+\mu_3)t} dt. \end{aligned}$$

Note that the hat symbol signifies this is an approximation of $p_L(l)$. Dividing and multiplying the expression by $(\mu_1 + \mu_3)^{2n_1+n_3-l}$, we obtain

$$\hat{p}_L(l) = \frac{\mu_1^{n_1+1} \mu_3^{n_1+n_3-l}}{(\mu_1 + \mu_3)^{2n_1+n_3-l} n_1!(n_1+n_3-l)!} \int_0^\infty ((\mu_1 + \mu_3)t)^{2n_1+n_3-l} e^{-(\mu_1+\mu_3)t} dt.$$

Now let $\tau = (\mu_1 + \mu_3)t$ and $N = 2n_1 + n_3 - l$. Then $dt = (\mu_1 + \mu_3)^{-1} d\tau$, and the integral in the above equation rewrites to

$$\int_0^\infty ((\mu_1 + \mu_3)t)^{2n_1+n_3-l} e^{-(\mu_1+\mu_3)t} dt = \frac{1}{\mu_1 + \mu_3} \int_0^\infty \tau^N e^{-\tau} d\tau.$$

The integral $\int_0^\infty \tau^N e^{-\tau} d\tau$ is known to be $N!$ [8]. Putting everything together, we have

$$\begin{aligned}\hat{p}_L(l) &= \frac{(2n_1 + n_3 - l)! \mu_1^{n_1+1} \mu_3^{n_1+n_3-l}}{n_1! (n_1 + n_3 - l)! (\mu_1 + \mu_3)^{2n_1+n_3-l+1}} \\ &= \binom{2n_1 + n_3 - l}{n_1} \left(\frac{\mu_1}{\mu_1 + \mu_3} \right)^{n_1+1} \left(\frac{\mu_3}{\mu_1 + \mu_3} \right)^{n_1+n_3-l}.\end{aligned}$$

There is a nice interpretation of this expression. During a time interval, $2n_1 + n_3 - l + 1$ events occur: there are $n_1 + 1$ services at S_1 and $2n_1 + n_3 - l$ services at S_3 . The $n_1 + 1$ power of $\frac{\mu_1}{\mu_1 + \mu_3}$ gives the probability that $n_1 + 1$ of these events are services at S_1 , and the $2n_1 + n_3 - l$ power of $\frac{\mu_3}{\mu_1 + \mu_3}$ gives the probability that $2n_1 + n_3 - l$ are services at S_3 . The binomial coefficient distributes the first n_1 services among the $2n_1 + n_3 - l$ events. Note that the last S_1 service to occur must be that of the tagged customer, hence its exclusion from the binomial coefficient.

The results of this subsection are summarised in the following lemma.

Lemma 6.1. *Suppose a customer joins the system and observes queue lengths $x_0 = (n_1, n_2, n_3, n_4)$. If she joins S_1 , her expected waiting time, based on an approximation which ignores queue 3 starvation, is*

$$\mathbb{E}(W|x_0, S_1) = \frac{n_1}{\mu_1} + \sum_{l=0}^{n_1+n_3} \binom{l}{\mu_3} \binom{2n_1 + n_3 - l}{n_1} \left(\frac{\mu_1}{\mu_1 + \mu_3} \right)^{n_1+1} \left(\frac{\mu_3}{\mu_1 + \mu_3} \right)^{n_1+n_3-l}.$$

The approximate expected waiting time if the customer goes to queue 2 is obtained equivalently. The approximate Nash equilibrium strategy is to choose a queue which gives a smaller value. If both queues give the same value, she may join queue 1 with probability p , and queue 2 with probability $1 - p$, where p is any number in $[0, 1]$.

We close this subsection by asking the question: when do we expect this approximation to perform well? Starvation occurs when S_3 is done serving all its customers, but S_1 still has customers to process. Therefore, we expect that when μ_1 is much faster than μ_3 , there is little starvation and the approximation performs well.

We also expect the approximation to perform well when μ_3 is much larger than μ_1 and n_3 is not large compared to n_1 . If this is the case, the tagged customer will rarely see any people at queue 3 when she is done with service at S_1 , because S_3 is much faster. Therefore the exact expected waiting time will be close to $\frac{n_1}{\mu_1}$. Lemma 6.1 provides the formula for the approximation of $E(W|x_0, S_1)$. We can remove the term where $l = 0$, as it does not affect the result. The approximation is then

$$\frac{n_1}{\mu_1} + \sum_{l=1}^{n_1+n_3} \binom{l}{\mu_3} \binom{2n_1 + n_3 - l}{n_1} \left(\frac{\mu_1}{\mu_1 + \mu_3} \right)^{n_1+1} \left(\frac{\mu_3}{\mu_1 + \mu_3} \right)^{n_1+n_3-l}.$$

Now suppose we fix μ_1 to some value and let μ_3 go to infinity. Then each term in the sum goes to 0, as $\frac{l}{\mu_3} \rightarrow 0$ for a positive l , $\frac{\mu_1}{\mu_1 + \mu_3} \rightarrow 0$, and $\frac{\mu_3}{\mu_1 + \mu_3}$ goes to 1. Thus, the

whole approximation goes to $\frac{n_1}{\mu_1}$, just as the true expectation does. We can also say when the approximation converges quickly to the true value in the case that μ_1 is fixed at some value and μ_3 goes to infinity. The higher n_1 is, the more quickly $\left(\frac{\mu_1}{\mu_1 + \mu_3}\right)^{n_1+1}$ goes to 0, and the approximation becomes more correct.

6.1.2. The Deterministic Approximation of the Nash Equilibrium Strategy

In this subsection we take the service time at each server S_i to be deterministic and equal to $1/\mu_i$. Again, suppose a customer arrives to the network, observes queue lengths (n_1, n_2, n_3, n_4) and joins queue 1. Assume further that there are no new arrivals after the tagged customer. We derive the expected waiting time $\mathbb{E}(W|(n_1, n_2, n_3, n_4), S_1)$, which can then be compared to the alternative $\mathbb{E}(W|(n_1, n_2, n_3, n_4), S_2)$ to determine the Nash equilibrium. We take as time step $\tau = \frac{1}{\mu_1}$ the length of a service at S_1 , and record at each time step the number of customers at stations 1 and 3.

Example 6.2. As a small illustrative example, let $\mu_1 = \mu_3 = 1, n_1 = 2, n_3 = 2$. Then, if the customer joins queue 1, the state of the $\{S_1, S_3\}$ tandem becomes $(3, 2)$. After one time step, one customer left queue 1 and proceeded to queue 3. One customer left queue 3 and departed from the system. The new state is $(2, 2)$. Continuing with the transitions, the system unfolds as follows

$$(3, 2) \rightarrow (2, 2) \rightarrow (1, 2) \rightarrow (0, 2) \rightarrow (0, 1) \rightarrow (0, 0).$$

The total time the customer was in the system is 5. During two of these time units the customer was in service, so the waiting time of the customer is 3.

The same logic can be applied to determine the waiting times in a more general setting. There are multiple cases to consider.

Case 1. $n_1 = n_3 = 0$.

If this is the case, the expected waiting time is 0 time units trivially, as both queues are empty.

Case 2. $\mu_1 \geq \mu_3, n_1 \geq 0$ and $n_3 \geq 1$.

The initial state of the queues is $(n_1 + 1, n_3)$, both of which are at least equal to 1. After one time step, there is one customer less in queue 1. This same customer joins queue 3. During this time step, $\lfloor \mu_3/\mu_1 \rfloor$ customers obtained service at S_3 . In total, there are now $n_3 + 1 - \lfloor \mu_3/\mu_1 \rfloor$ customers in queue 3. This quantity rewrites to $\lceil n_3 + 1 - \mu_3/\mu_1 \rceil$. The dynamics of these queues will stay the same, as long as neither queue depletes. At each time step, the difference in length of queue 3 is $1 - \lfloor \mu_3/\mu_1 \rfloor \geq 0$. As $\mu_1 \geq \mu_3$, this difference is positive, meaning that the queue length either grew or stayed the same. On the other hand, queue 1 loses one customer at each time step. Therefore, queue 1 depletes before queue 3, and we have the following transitions

$$(n_1 + 1, n_3) \rightarrow \left(n_1, \left\lceil n_3 + 1 - \frac{\mu_3}{\mu_1} \right\rceil\right) \rightarrow \left(n_1 - 1, \left\lceil n_3 + 2 - 2\frac{\mu_3}{\mu_1} \right\rceil\right) \rightarrow \dots$$

$$\rightarrow \left(1, \left\lceil n_3 + n_1 - n_1 \frac{\mu_3}{\mu_1} \right\rceil\right) \rightarrow \left(0, \left\lceil n_3 + n_1 + 1 - (n_1 + 1) \frac{\mu_3}{\mu_1} \right\rceil\right).$$

The last transition occurs at the time step $n + 1$, which happens at time $(n + 1)/\mu_1$.

After this, queue 3 decreases by $\lfloor \mu_3/\mu_1 \rfloor$ customers at each time step. The transitions are then

$$\left(0, \left\lceil n_3 + n_1 + 1 - (n_1 + 1) \frac{\mu_3}{\mu_1} \right\rceil\right) \rightarrow \left(0, \left\lceil n_3 + n_1 + 1 - (n_1 + 2) \frac{\mu_3}{\mu_1} \right\rceil\right) \rightarrow \dots$$

$$\dots \rightarrow \left(0, \left\lceil n_3 + n_1 + 1 - (n_1 + 1 + k) \frac{\mu_3}{\mu_1} \right\rceil\right).$$

We are looking for k such that $n_3 + n_1 + 1 - (n_1 + 1 + k) \frac{\mu_3}{\mu_1} = 0$. Note that k does not have to be a natural number here: the last service may finish before a full time step. The equation which determines k rewrites to

$$n_1 + k + 1 = \frac{\mu_1}{\mu_3}(n_1 + n_3 + 1),$$

which gives

$$k = \frac{\mu_1}{\mu_3}(n_1 + n_3 + 1) - (n_1 + 1).$$

These k time steps take $\frac{(n_1+n_3+1)}{\mu_3} - \frac{n_1+1}{\mu_1}$ time. In total, it takes $\frac{(n_1+n_3+1)}{\mu_3}$ time for the system to clear. This time is the customers sojourn time. Thus, the customer's waiting time is $\frac{(n_1+n_3)}{\mu_3} - \frac{1}{\mu_1}$.

Case 3. $\mu_1 \geq \mu_3$, $n_1 \geq 1$ and $n_3 = 0$,

If this is the case, the initial state of the queues is $(n_1 + 1, 0)$, and after one time step the state is $(n_1, 1)$. This state is of the same form as the initial state in the previous case, which means we can use the results of that case, if we make the substitution $n_3 \rightarrow 1, n_1 \rightarrow n_1 - 1$. The customer's waiting time after this initial time step is $\frac{(n_1-1)+1}{\mu_3} - \frac{1}{\mu_1}$. Adding the initial time step, the customer's total waiting time is $\frac{n_1}{\mu_3}$.

Next, we consider the cases where the second stage server is quicker than the first stage one.

Case 4. $\mu_1 < \mu_3$, $n_1 \geq 0$ and $n_3 \geq 1$.

Again, at each time step the number of customers decreases by 1 at S_1 , and changes by $1 - \lfloor \frac{\mu_3}{\mu_1} \rfloor$ at S_3 , only now this change is negative, and queue 3 starts losing customers right from the beginning. Once this queue depletes, at each subsequent time step it will be empty again: if there are still customers at S_1 , one customer per time unit will enter queue 3, she will go straight into service at S_3 , and will obtain service before the end of the time step. Let us assume that after k time steps, neither queue has depleted yet.

Then, the queue lengths are $\left(n_1 + 1 - k, \left\lceil n_3 + k \left(1 - \frac{\mu_3}{\mu_1}\right) \right\rceil\right)$. We split the proof into cases based on which queue depletes first.

Subcase (i): Suppose $\left\lceil n_3 + (n_1 + 1) \left(1 - \frac{\mu_3}{\mu_1}\right) \right\rceil > 0$.

After $n_1 + 1$ time steps, there are 0 people in queue 1. Suppose for the sake of contradiction that queue 3 depleted before queue 1, say at time step l , so that $n_3 + l \left(1 - \left\lfloor \frac{\mu_3}{\mu_1} \right\rfloor\right)$ is 0. But $l < n_1 + 1$ and $1 - \left\lfloor \frac{\mu_3}{\mu_1} \right\rfloor < 0$, so

$$0 = n_3 + l \left(1 - \left\lfloor \frac{\mu_3}{\mu_1} \right\rfloor\right) > n_3 + (n_1 + 1) \left(1 - \left\lfloor \frac{\mu_3}{\mu_1} \right\rfloor\right) > 0.$$

Therefore $0 > 0$, and we have a contradiction. Queue 3 has not depleted at time $n_1 + 1$, and at that time it has $n_3 + (n_1 + 1) \left(1 - \left\lfloor \frac{\mu_3}{\mu_1} \right\rfloor\right)$ customers. The rest of the proof is now identical to the $\mu_1 \geq \mu_3$ case: it takes another $k = \frac{\mu_1}{\mu_3}(n_1 + n_3 + 1) - (n_1 + 1)$ time steps for the queue to empty. In total, the waiting time of the customer is $\frac{(n_1 + n_3)}{\mu_3} - \frac{1}{\mu_1}$.

Subcase (ii): Suppose $\left\lceil n_3 + (n_1 + 1) \left(1 - \frac{\mu_3}{\mu_1}\right) \right\rceil \leq 0$.

This is the case where queue 3 empties before, or at the same time as queue 1. As previously mentioned, once queue 3 empties, it receives one customer per time unit, and processes that customer in the same time unit, so that no customer has to wait for service at S_3 . Therefore, the total time until the system empties is the time it takes for all customers to receive service at S_1 plus the time it takes for the last arriving customer to receive service at S_3 . Thus, the sojourn time of the tagged customer is $\frac{n_1 + 1}{\mu_1} + \frac{1}{\mu_3}$, and the customer's waiting time is $\frac{n_1 + 1}{\mu_1} + \frac{1}{\mu_3} - \frac{1}{\mu_1} - \frac{1}{\mu_3} = \frac{n_1}{\mu_1}$.

Case 5. $\mu_1 < \mu_3, n_1 \geq 1$ and $n_3 = 0$.

After one time step, the state is $(n_1, 1)$ - of the same form as the initial state in the previous case. Thus, we make the substitution $n_3 \rightarrow 1, n_1 \rightarrow n_1 - 1$ and use the results of the previous case. The solution branches into two subcases, depending on the sign of $\left\lceil 1 + n_1 \left(1 - \frac{\mu_3}{\mu_1}\right) \right\rceil$: if this quantity is positive, the waiting time (after the initial time step) is $\frac{n_1}{\mu_3} - \frac{1}{\mu_1}$; otherwise it is $\frac{n_1 - 1}{\mu_1}$. Adding the initial time step to this waiting time, the solution is $\frac{n_1}{\mu_3}$ if $\left\lceil 1 + n_1 \left(1 - \frac{\mu_3}{\mu_1}\right) \right\rceil > 0$, and $\frac{n_1}{\mu_1}$ otherwise.

The results of this subsection are summarised in the following lemma.

Lemma 6.3. *Suppose a customer observes state $x_0 = (n_1, n_2, n_3, n_4)$ upon joining the network. If $\mu_1 \geq \mu_3$, her expected waiting time based on the deterministic approximation, should she join queue 1, is*

$$\mathbb{E}(W|x_0, S_1) = \begin{cases} \frac{n_1}{\mu_3} & \text{if } n_3 = 0 \\ \frac{n_1 + n_3}{\mu_3} - \frac{1}{\mu_1} & \text{else.} \end{cases}$$

If $\mu_1 < \mu_3$, her expected waiting time is

$$\mathbb{E}(W|x_0, S_1) = \begin{cases} \frac{n_1}{\mu_1} & \text{if } n_3 \geq 1 \text{ and } A \text{ or } n_1 \geq 1, n_3 = 0 \text{ and } B, \\ \frac{n_1}{\mu_3} & \text{if } n_1 \geq 1, n_3 = 0 \text{ and } \left\lceil 1 + n_1 \left(1 - \frac{\mu_3}{\mu_1}\right) \right\rceil > 0, \\ \frac{(n_1 + n_3)}{\mu_3} - \frac{1}{\mu_1} & \text{else,} \end{cases}$$

where $A = \left\{ \left\lceil n_3 + (n_1 + 1) \left(1 - \frac{\mu_3}{\mu_1}\right) \right\rceil \leq 0 \right\}$ and $B = \left\{ \left\lceil 1 + n_1 \left(1 - \frac{\mu_3}{\mu_1}\right) \right\rceil \leq 0 \right\}$. The approximate expected waiting time if the customer goes to queue 2 is obtained equivalently. The deterministic approximation of a Nash equilibrium strategy is to choose the server with the smaller expected waiting time. If both queues give the same value, the customer may join queue 1 with probability p , and queue 2 with probability $1 - p$, where p is any number in $[0, 1]$.

We expect this approximation to perform well in a busy system, as deterministic approximations of queues generally perform better in systems with a heavy load.

6.1.3. The Exact Nash Equilibrium Strategy

We will now derive the equations that characterise the Nash equilibrium strategy in the simplified network. The equations are exact in that we do not make any approximations to derive them. However, we obtain a set of equations for each possible initial joining state, of which there are infinitely many. In practice, we can only solve the equations for a finite number of initial joining states.

Suppose a customer sees state $x_0 = (n_1^0, n_2^0, n_3^0, n_4^0)$ upon arrival, where the 0 superscript signifies that this is the initial state of the network before joining any queue. We will provide a way to determine the expected waiting time of this customer if the customer joins queue 1, and the expected waiting time if the customer joins queue 2. We will call these values $\mathbb{E}(W|x_0, S_1)$ and $\mathbb{E}(W|x_0, S_2)$, respectively. Then the Nash equilibrium strategy is to simply join the queue which gives a smaller value. We will assume that once a customer has joined one of the queues, there are no new arrivals. As explained at the start of this chapter, the waiting times are not affected by this assumption.

Let $X(n_1, n_3)$ be the expected clearing time of the $\{S_1, S_3\}$ tandem if there are currently n_1 customers at queue 1, and n_3 customers at queue 3. This is the time it takes for the system to empty. Then $X(n_1^0 + 1, n_3^0)$ is also precisely the expected sojourn time of the tagged customer. This is because the tagged customer will be the last to leave the $\{S_1, S_3\}$ tandem of queues, as there are no new arrivals after her and no possibility of overtaking. Note that we only use subscripts n_1, n_3 ; this is because once the tagged customer has joined queue 1, her expected waiting time is not affected by the behaviour of the tandem $\{S_2, S_4\}$. The only relevant events are services at S_1 and at S_3 . Therefore, if $1 \leq n_1 \leq n_1^0 + 1$ and $1 \leq n_3 \leq n_1^0 + n_3^0 + 1$, the following recursive relation holds

$$X(n_1, n_3) = \frac{1}{\mu_1 + \mu_3} + \frac{\mu_1}{\mu_1 + \mu_3} X(n_1 - 1, n_3 + 1) + \frac{\mu_3}{\mu_1 + \mu_3} X(n_1, n_3 - 1).$$

The first term on the right hand side is the expected time until an event occurs. The time until an event occurs is exponentially distributed with rate $\mu_1 + \mu_3$. In case of an event, it is a service at S_1 with probability $\frac{\mu_1}{\mu_1 + \mu_3}$, in which case there are $n_1 - 1$ customers in queue 1 and $n_3 + 1$ customers at queue 3. The event is a service at S_3 with probability $\frac{\mu_3}{\mu_1 + \mu_3}$, in which case there are n_1 customers in queue 1 and $n_3 - 1$ customers at queue 3. The bound $1 \leq n_1 \leq n_1^0 + 1$ comes from the fact that the queue length at S_1 steadily decreases from $n_1^0 + 1$ to 0 during the sojourn time of the tagged customer. On the other hand, the queue length at S_3 can increase up to $n_1^0 + n_3^0 + 1$, if all the services at S_1 occur before any services at S_3 . Therefore the queue length at S_3 is bounded by 0 from below, and by $n_1^0 + n_3^0 + 1$ from above.

On the boundary, where not all services are possible, we have for $0 \leq n_1 \leq n_1^0 + 1$

$$X(0, n_3) = \frac{1}{\mu_3} + X(0, n_3 - 1),$$

and for $0 \leq n_3 \leq n_1^0 + n_3^0 + 1$

$$X(n_1, 0) = \frac{1}{\mu_1} + X(n_1 - 1, 1).$$

Finally, when the system is empty, $X(0, 0) = 0$ holds trivially. We were not able to obtain a closed form solution for $X(n_1, n_3)$. Instead, we obtained the results numerically as a solution to a linear system of equations. Subtracting the expected service duration at servers S_1 and S_3 gives the expected waiting time of the tagged customer, $E(W|x_0, S_1) = X(n_1^0 + 1, n_3^0) - \frac{1}{\mu_1} - \frac{1}{\mu_3}$.

Using the same logic, we can define $Y(n_2, n_4)$ to be the expected clearing time of the $\{S_2, S_4\}$ tandem when there are n_2 customers in queue 2 and n_4 customers at queue 4. Then $E(W|x_0, S_2) = X(n_2^0 + 1, n_4) - \frac{1}{\mu_2} - \frac{1}{\mu_4}$, and Y is characterised by the following equation when $1 \leq n_2 \leq n_2^0 + 1$ and $1 \leq n_4 \leq n_2^0 + n_4^0 + 1$

$$Y(n_2, n_4) = \frac{1}{\mu_2 + \mu_4} + \frac{\mu_2}{\mu_2 + \mu_4} Y(n_2 - 1, n_4 + 1) + \frac{\mu_4}{\mu_2 + \mu_4} Y(n_2, n_4 - 1).$$

When $0 \leq n_2 \leq n_2^0 + 1$

$$Y(0, n_4) = \frac{1}{\mu_4} + Y(0, n_4 - 1),$$

and when $0 \leq n_4 \leq n_2^0 + n_4^0 + 1$

$$Y(n_2, 0) = \frac{1}{\mu_2} + Y(n_2 - 1, 1).$$

6.2. The Full Model - Nash

In this subsection we present an algorithm which finds the Nash equilibrium strategy for the full model. We solve the problem using policy updates. We assume an initial

joining policy to be the weighted join shortest queue policy with queues 1 and 2. This means that customers only look at the length of queue 1 and queue 2 and choose to join the queue with the smallest expected waiting time.

Based on this joining strategy, we will set up and solve two systems of linear equations. The first system contains, for each state a customer may observe upon arrival, equations in the customer's expected waiting time, should the customer choose to join queue 1. The second system of equations is equivalent, only now we assume the customer joins queue 2. We solve these two systems numerically. We then obtain for each initial state an approximation of the waiting time of a customer if the customer goes to queue 1, and the approximate expected waiting time if the customer goes to queue 2. For each state, we can compare these two quantities, choose the smaller one, and set this as the new joining policy. We continue with this policy iteration, until the policy no longer changes.

We will now formalise the first step of the policy iteration. To do this, the following definition is needed.

Definition 6.4. Let $X(n_1, n_2, n_3, n_4, n_5, i)$ be the expected remaining waiting time of a tagged customer if she is in queue 1, her distance to the end of queue 1 is i and the state she currently observes is $(n_1, n_2, n_3, n_4, n_5)$. Equivalently, let $Y(n_1, n_2, n_3, n_4, n_5, i)$ be the expected remaining waiting time of a tagged customer if she is in queue 2, her distance to the end of queue 2 is i and the state she currently observes is $(n_1, n_2, n_3, n_4, n_5)$.

Remark. The state a customer observes also includes the customer. For example, the notation $X(n_1, n_2, n_3, n_4, n_5, i)$ means that there are n_1 people in queue 1, and the tagged customer is one of them. By “distance to end of queue 1”, we mean the number of people behind the tagged customer in queue 1.

Unlike in the simplified network, overtaking is possible in the full network, so the Nash strategy depends on the arrival rate. This means that we cannot assume there are no arrivals after the tagged customer, as we did in the previous sections. But if this is the case, we need to be able to keep track of where the tagged customer is in the queue. Hence, we also include the tagged customer's distance to the end of the queue in the state description.

Suppose a customer has just arrived to the system, she has not joined a queue yet, and she observes the queue lengths $(n_1, n_2, n_3, n_4, n_5)$. She will compare $X(n_1 + 1, n_2, n_3, n_4, n_5, 0)$ and $Y(n_1, n_2 + 1, n_3, n_4, n_5, 0)$ and choose the queue which gives a smaller expected waiting time - this is the Nash equilibrium. For simplicity we assume that in case of ties, she will go to S_1 . We will now derive the equations that characterise X and Y .

Let $(n_1, n_2, n_3, n_4, n_5, i)$ be such that $n_i \in \mathbb{Z}_{\geq 0}$ and $0 \leq i < n_1$. Suppose for the moment that all n_i are greater than 0. This way, a service is possible at each of the servers, and arrivals can still happen. Let $p_0 = \frac{\lambda}{\lambda + \sum_{i=1}^5 \mu_i}$, $p_j = \frac{\mu_j}{\lambda + \sum_{i=1}^5 \mu_i}$ for $j =$

1, 2, 3, 4, 5. We then have the following equation

$$\begin{aligned}
X(n_1, n_2, n_3, n_4, n_5, i) = & \frac{1}{\lambda + \sum_{i=1}^5 \mu_i} + p_1 X(n_1 - 1, n_2, \tilde{n}_3, \tilde{n}_4, n_5, i) \\
& + p_2 X(n_1, n_2 - 1, n_3, \hat{n}_4, \hat{n}_5, i) \\
& + p_3 X(n_1, n_2, n_3 - 1, n_4, n_5, i) + p_4 X(n_1, n_2, n_3, n_4 - 1, n_5, i) \\
& + p_5 X(n_1, n_2, n_3, n_4, n_5 - 1, i) \\
& + p_0 X(n_1 + 1, n_2, n_3, n_4, n_5, i + 1) \mathbb{P}(\text{"new arrival joins queue 1"}) \\
& + p_0 X(n_1, n_2 + 1, n_3, n_4, n_5, i) \mathbb{P}(\text{"new arrival joins queue 2"}).
\end{aligned} \tag{6.2}$$

The first term is the expected time until the next event. The other two terms in the first two rows are the transitions when there is a service at S_1 or S_2 . Customers who finished service at a first stage server will use the weighted join shortest queue strategy to decide between the two available servers in the second stage. Thus,

$$\begin{aligned}
(\tilde{n}_3, \tilde{n}_4) &= \begin{cases} (n_3 + 1, n_4) & \frac{n_3}{\mu_3} \leq \frac{n_4}{\mu_4}, \\ (n_3, n_4 + 1) & \text{else,} \end{cases} \\
(\hat{n}_4, \hat{n}_5) &= \begin{cases} (n_4 + 1, n_5) & \frac{n_4}{\mu_4} < \frac{n_5}{\mu_5}, \\ (n_4, n_5 + 1) & \text{else.} \end{cases}
\end{aligned}$$

The third and fourth row are the transitions when there is a service at the second stage servers. The last two rows are the transitions when there is a new arrival. This new arrival sees state $(n_1, n_2, n_3, n_4, n_5)$ and she chooses whether to join queue 1 or 2. The probability she goes to queue 1 is the following

$$\mathbb{P}(\text{"new arr. goes to } S_1") = \begin{cases} 1 & \text{if } X(n_1 + 1, n_2, n_3, n_4, n_5, 0) \leq Y(n_1, n_2 + 1, n_3, n_4, n_5, 0), \\ 0 & \text{else} \end{cases}$$

because if a new customer arrives, she will see state $(n_1, n_2, n_3, n_4, n_5)$. She will join queue 1 if her expected waiting time is less or equal than her expected waiting time if she joins queue 2. Her expected waiting time if she joins queue 1 is $X(n_1 + 1, n_2, n_3, n_4, n_5, 0)$, because including her, there are $n_1 + 1$ people at queue 1, and she is the last to join, so $i = 0$. Similarly, if she joins queue 2 her expected waiting time is $Y(n_1, n_2 + 1, n_3, n_4, n_5, 0)$. In the first step of the policy iteration, we assume the joining strategy is simply weighted join shortest queue with queues 1 and 2, so that

$$\mathbb{P}(\text{"new arrival goes to } S_1") = \begin{cases} 1 & \text{if } \frac{n_1}{\mu_1} \leq \frac{n_2}{\mu_2}, \\ 0 & \text{else.} \end{cases}$$

Therefore, Equation (6.2) is simply a linear equation. We can set up similar equations also when some stations are idle. In that case, let us define the indicator variables $I_i = \mathbb{1}_{\{n_i \geq 0\}}$ for i in $\{1, 2, 3, 4, 5\}$. We can then simply add these indicator variables

to Equation (6.2), so that if a station is idle it does not contribute to the transitions. We redefine the p_j so they include the indicator variables. Let $p_0 = \frac{\lambda}{\lambda + \sum_{i=1}^5 \mu_i I_i}$, $p_j = \frac{\mu_j I_j}{\lambda + \sum_{i=1}^5 \mu_i I_i}$ for $j = 1, 2, 3, 4, 5$. We then have the following equation

$$\begin{aligned}
X(n_1, n_2, n_3, n_4, n_5, i) = & \frac{1}{\lambda + \sum_{i=1}^5 \mu_i I_i} + p_1 X(n_1 - 1, n_2, \tilde{n}_3, \tilde{n}_4, n_5, i) \\
& + p_2 X(n_1, n_2 - 1, n_3, \hat{n}_4, \hat{n}_5, i) \\
& + p_3 X(n_1, n_2, n_3 - 1, n_4, n_5, i) + p_4 X(n_1, n_2, n_3, n_4 - 1, n_5, i) \\
& + p_5 X(n_1, n_2, n_3, n_4, n_5 - 1, i) \\
& + p_0 X(n_1 + 1, n_2, n_3, n_4, n_5, i + 1) \mathbb{P}(\text{"new arrival joins queue 1"}) \\
& + p_0 X(n_1, n_2 + 1, n_3, n_4, n_5, i) \mathbb{P}(\text{"new arrival joins queue 2"}).
\end{aligned} \tag{6.3}$$

Similarly,

$$\begin{aligned}
Y(n_1, n_2, n_3, n_4, n_5, i) = & \frac{1}{\lambda + \sum_{i=1}^5 \mu_i I_i} + p_1 Y(n_1 - 1, n_2, \tilde{n}_3, \tilde{n}_4, n_5, i) \\
& + p_2 Y(n_1, n_2 - 1, n_3, \hat{n}_4, \hat{n}_5, i) \\
& + p_3 Y(n_1, n_2, n_3 - 1, n_4, n_5, i) + p_4 Y(n_1, n_2, n_3, n_4 - 1, n_5, i) \\
& + p_5 Y(n_1, n_2, n_3, n_4, n_5 - 1, i) \\
& + p_0 Y(n_1 + 1, n_2, n_3, n_4, n_5, i) \mathbb{P}(\text{"new arrival joins queue 1"}) \\
& + p_0 Y(n_1, n_2 + 1, n_3, n_4, n_5, i + 1) \mathbb{P}(\text{"new arrival joins queue 2"}).
\end{aligned} \tag{6.4}$$

Combining these equations, we get two systems of equations of the form $Ax + b = x$, or equivalently $Mx = b$, with $M = I - A$. Since the state space is unbounded, we must truncate it to some appropriate size before we can numerically solve the system. We truncate the state space to $[N_1, N_2, N_3, N_4, N_5]$, where N_i is the maximal queue length we obtain in a simulation in which the joining policy is weighted join shortest queue with queues 1 and 2. In states where queue i reaches length N_i , we assume that customers balk.

That completes the first step of the policy iteration. Once X and Y are obtained, the individually optimal strategy is to choose the smaller value, and choose either with some probability if the values are equal. The results is a new joining policy π_1 . We can then set up the same equations as Equation (6.3) and (6.4). The only difference is that $\mathbb{P}(\text{"new arrival joins queue 1"})$ and $\mathbb{P}(\text{"new arrival joins queue 2"})$ are now determined not by the WJSQ policy, but by π_1 . Solving the system results in a new policy, and the same steps are repeated. We stop the iteration when the policy no longer changes. Then the resulting policy is the approximate Nash equilibrium. This procedure is summarised in the following algorithm.

Algorithm 1 Nash Policy Iteration

Initialise $s = \text{WJSQ}$ on S_1 and S_2

repeat

1. Using s as the joining policy, define systems of linear equations in $X(n_1, n_2, n_3, n_4, n_5, i)$ and $Y(n_1, n_2, n_3, n_4, n_5, i)$ as in Equations (6.3) and (6.4).
2. Solve the two systems.
3. Compute the updated policy:

$$s' = \begin{cases} 1 & \text{if } X(n_1 + 1, n_2, n_3, n_4, n_5, 0) \leq Y(n_1, n_2 + 1, n_3, n_4, n_5, 0) \\ 2 & \text{else.} \end{cases}$$

4. Set $s = s'$.

until Policy no longer changes

return approximate Nash strategy s_N

7. Fully Observable Case - Numerical Observations

In this chapter we present some numerical results regarding the observable case. First, results on the simplified network are presented in Section 7.1. Figures illustrating the accuracy of the approximations are shown, as well results obtained from a simulation of the network which compare the various strategies in terms of how well they perform socially. After this, results pertaining to the full network are discussed in Section 7.2. First, the influence of the arrival and service rates on the strategy is studied. This is followed by some simulation results where the full network strategies are compared in terms of social cost.

7.1. The Simplified Network

In this section we analyse the exact Nash strategy and the corresponding approximations in the simplified network setting. We compare the approximations with the exact Nash solution for various parameter values and initial queue lengths. We also showcase how well the approximations and the exact Nash strategy perform in a simulation, with respect to the average waiting time during a simulation.

7.1.1. Comparison of the No Starvation Approximation with the Exact Nash Equilibrium

We compare the “no starvation” approximation of the Nash strategy with the exact Nash strategy. We set all network parameters (the arrival rate and the service rates) and n_3 and n_4 to some values. We then let n_1 and n_2 be in some range, and we plot for each point (n_1, n_2) the best response, according to exact Nash and the approximation. Figure 7.1 shows the best response when all rates are equal to 1, $n_3 = n_4 = 0$ and n_1 and n_2 are both in the range $[0, 40]$.

For each point (n_1, n_2) , the colour of the top half circle states what the exact Nash strategy prescribes in that state, while the colour of the bottom half circle informs us what the no starvation Nash approximation strategy prescribes. When a half circle is coloured blue, it indicates that the expected waiting times are equal whether one joins queue 1 or queue 2. Unsurprisingly, there is a line of blue circles along the diagonal. For both the exact Nash, and the no starvation approximation, when the situation is symmetrical, the computed expected waiting times are equal. As we can see from the colours matching at each circle, the two strategies agree in this case.

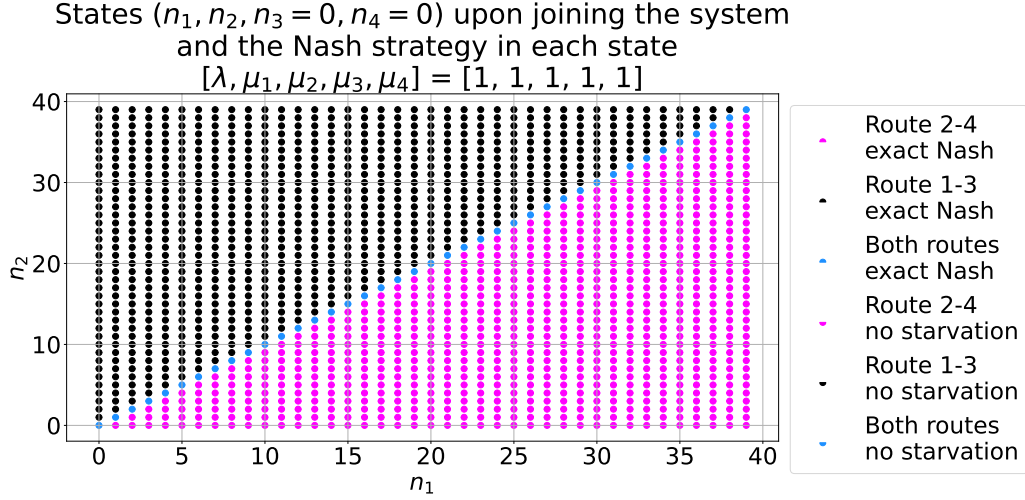


Figure 7.1: The two strategies agree at all points when the network is completely symmetric.

Figure 7.2 shows the strategies when all rates are equal to 1, but $n_3 = 10, n_4 = 0$. There are now more points where both strategies prescribe joining queue 2, as the $\{S_1, S_3\}$ tandem is more congested. We also see that the two strategies do not always agree. There is a diagonal-like boundary where the strategies differ, but once further from the diagonal, the choice becomes more apparent and the strategies agree again.

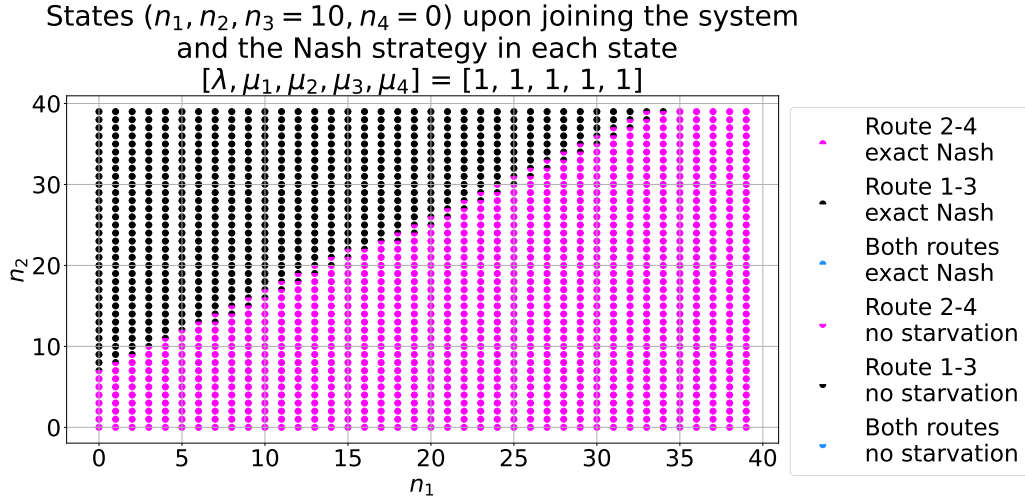


Figure 7.2: The strategies differ at 60 points when the initial distribution of customers is not equal in both tandems and the service speed is equal in both stages.

Figure 7.3 shows the two strategies when the first stage servers are 2 times faster than the second stage servers, $n_3 = 10$ and $n_4 = 0$. Notice that there are more pink circles

than in the previous figure, i.e. both strategies recommend route 2-4 more often. This is because the second stage servers are slower, meaning that an initially high value of n_3 is more detrimental. More importantly, the strategies agree on all points in the plot. We conjectured that the no starvation approximation will perform well when the first stage servers are much faster than the second stage servers; it seems that this conjecture is correct.

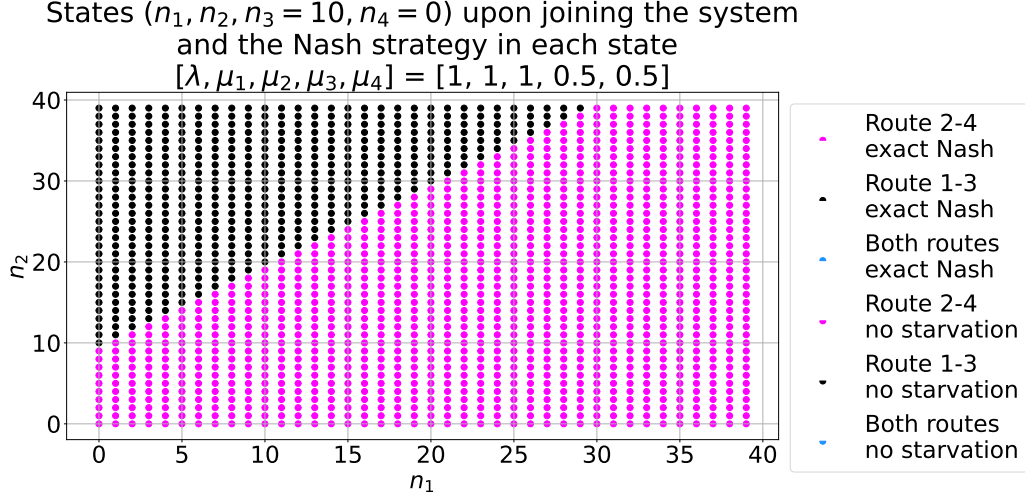


Figure 7.3: The strategies agree at all points when the initial distribution of customers is not equal in both tandems and first stage servers are faster than second stage servers.

The same effect can be observed in Figure 7.4. Here $n_3 = 10, n_4 = 0$ and the second stage servers are two times faster than the first stage servers. The strategies agree at all but four points. These four points are $(3, 5), (5, 7), (9, 10), (10, 11)$, where n_1 and n_2 are relatively small. This is in line with our conjecture that the approximation will perform better when the first stage servers have longer queues.

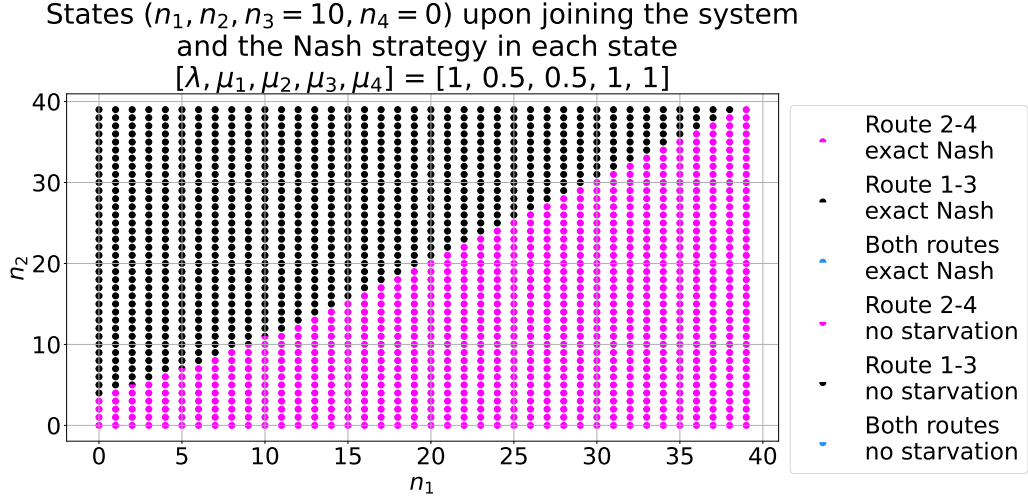


Figure 7.4: The strategies differ at 4 points when the initial distribution of customers is not equal in both tandems and first stage servers are slower than second stage servers.

Given values of n_1, n_3, μ_1 and μ_3 , we can calculate the expected waiting time for a customer who goes up to S_1 . For the same customer, we can also compute the approximate expected waiting time using the no starvation approximation. We can then define the absolute error of the no starvation approximation to be the absolute value of the difference of these two values. Figure 7.5 shows the absolute error of the no starvation approximation, as a function of μ_3 . Here μ_1 is set to 1, n_3 is set to 10, and μ_3 varies from 1 to 20. Five different curves are plotted, for $n_1 \in \{1, 2, 3, 4, 5\}$. Each curve tends to 0 as μ_3 grows, but more quickly when n_1 is large.

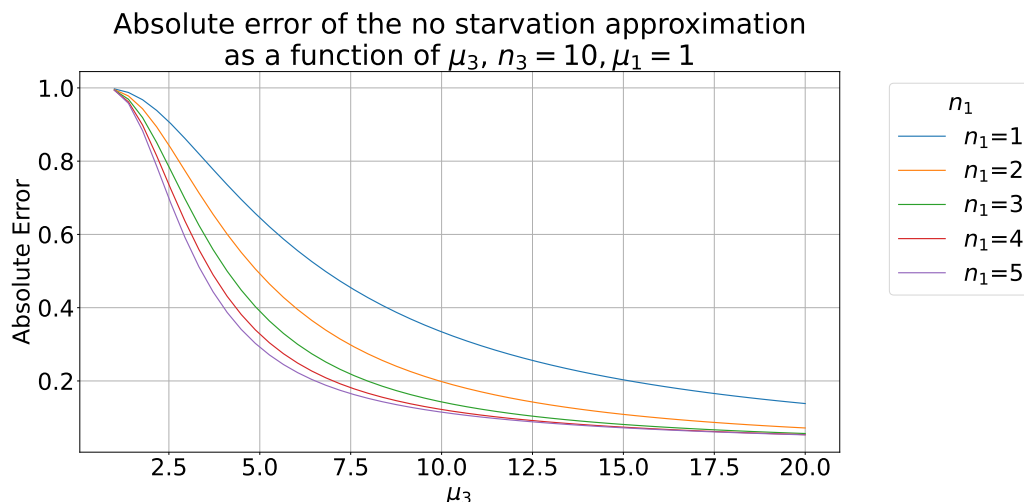


Figure 7.5: The no starvation approximation accuracy depends positively on both n_1 and μ_3/μ_1 .

7.1.2. Comparison of the Deterministic Approximation with the Exact Nash Equilibrium

In this subsection we investigate how well the deterministic approximation of the Nash strategy performs against the exact Nash strategy. We make plots of the same type as in the previous subsection. Figure 7.6 shows the two strategies when all network parameters are equal to 1 and $n_3 = n_4 = 0$. The strategies agree on all points in the plot, and there is a diagonal boundary where both responses are optimal. Similarly to the no starvation approximation, the deterministic approximation performs well in a symmetric setting.

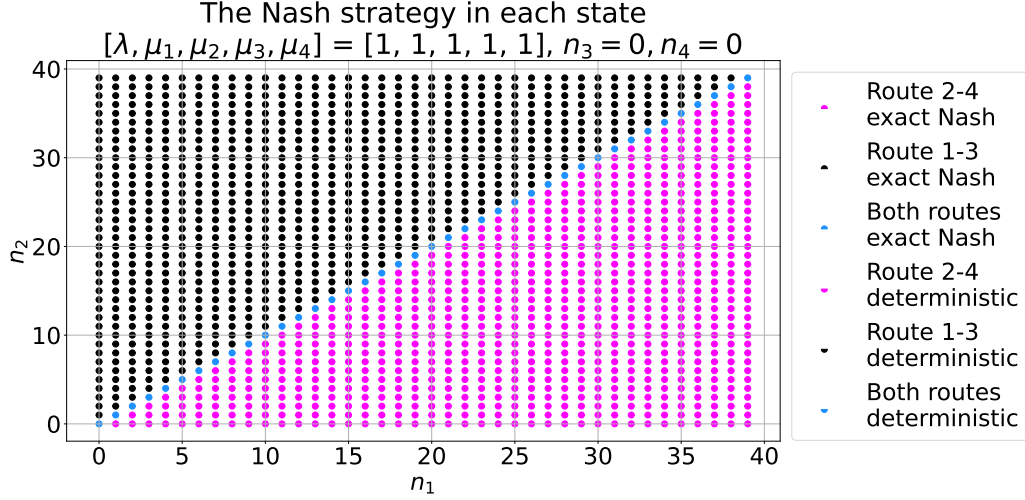


Figure 7.6: The two strategies agree at all points when the network is completely symmetric.

Figure 7.7 shows the two strategies when all network parameters are equal to 1, $n_3 = 10$ and $n_4 = 0$. The two strategies disagree at a total of 140 points, with the deterministic approximation prescribing route 2-4 more often than the exact Nash strategy.

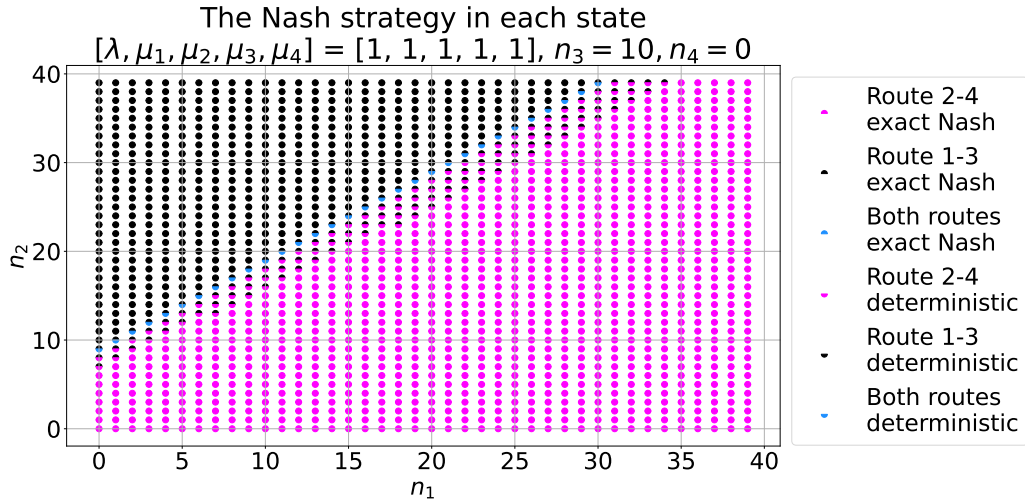


Figure 7.7: The two strategies disagree at 140 points when queues 3 and 4 are reasonably short.

On the other hand, when we increase the number of customers in the second stage queues, the strategies agree more often, as Figure 7.8 shows. Here again, all the network parameters are equal to 1. This could suggest that if the queues are long, the deterministic approximation performs well. However, Figure 7.9 contradicts this statement.

Here, all network parameters are equal to 1, $n_3 = 1$ and $n_4 = 0$. Here, the queue lengths are smaller than in Figure 7.7, and yet the approximation performs better. It remains unclear exactly how the initial queue lengths influence the approximation performance.

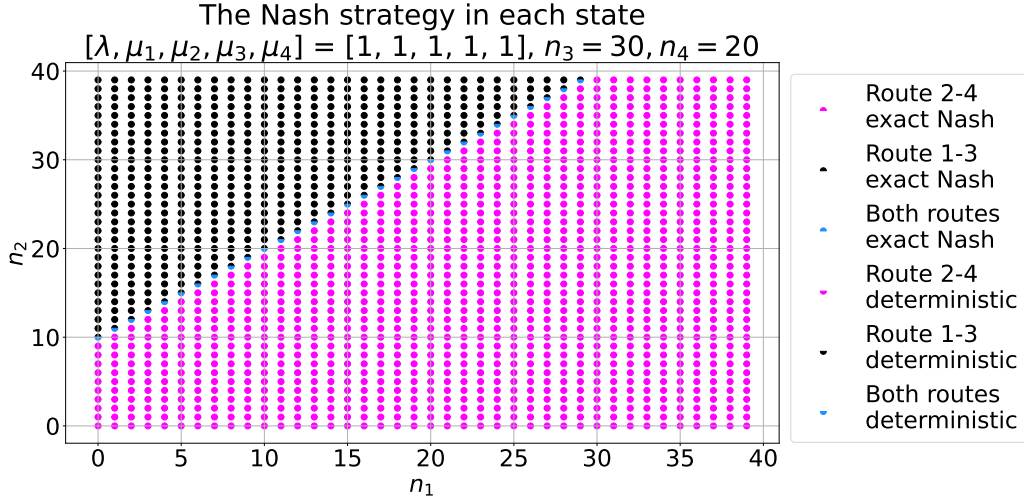


Figure 7.8: The two strategies disagree at 30 points when queue 3 and 4 are long.

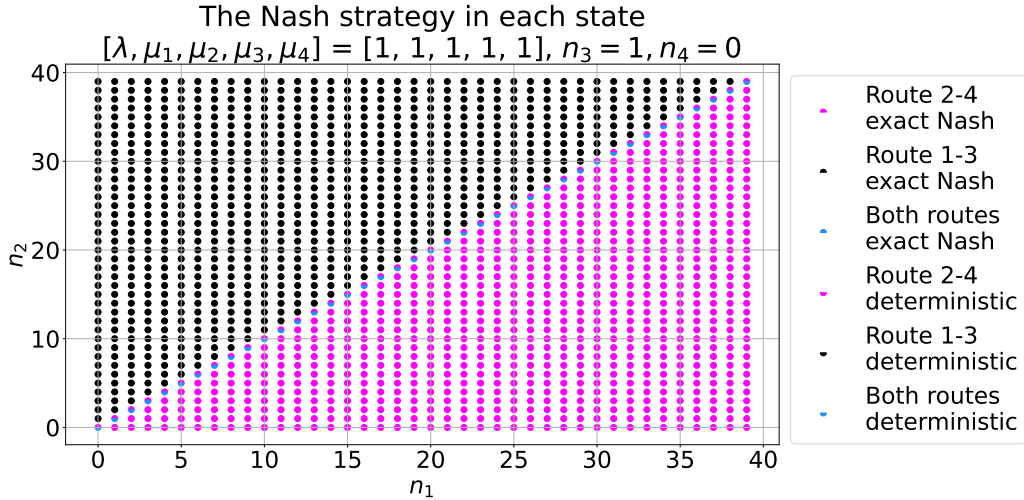


Figure 7.9: The two strategies disagree at 40 points when queue 3 and 4 are short.

The influence of the load on performance is much easier to see. The effect is clearly shown in Figure 7.10. Here, the service rates are all fixed at 1, and the arrival rate ranged from 0.1 to 1.9. For each value of λ , we perform a simulation of length 500 000 time units with a warm up time of 10 000 time units. Customers in the simulation follow the Nash strategy. During the simulation, we record for each joining state the Nash decision

and the deterministic approximation decision. We calculate the proportion of decisions where the exact Nash and deterministic strategy agree. This proportion is taken as a measure of accuracy of the approximation. We plot the resulting proportions against the arrival rates. It is clear that the higher the arrival rate, and therefore the system load, the higher the accuracy of the deterministic approximation is. Note that when λ is near 0, the strategies agree only 60 % of the time. This number is surprisingly small, but could perhaps be explained by the fact that the customers follow a Nash strategy. The Nash strategy will keep the queue lengths somewhat balanced. However, when the queue lengths are such that the waiting times at both tandems are almost equal, the deterministic strategy does not perform well.

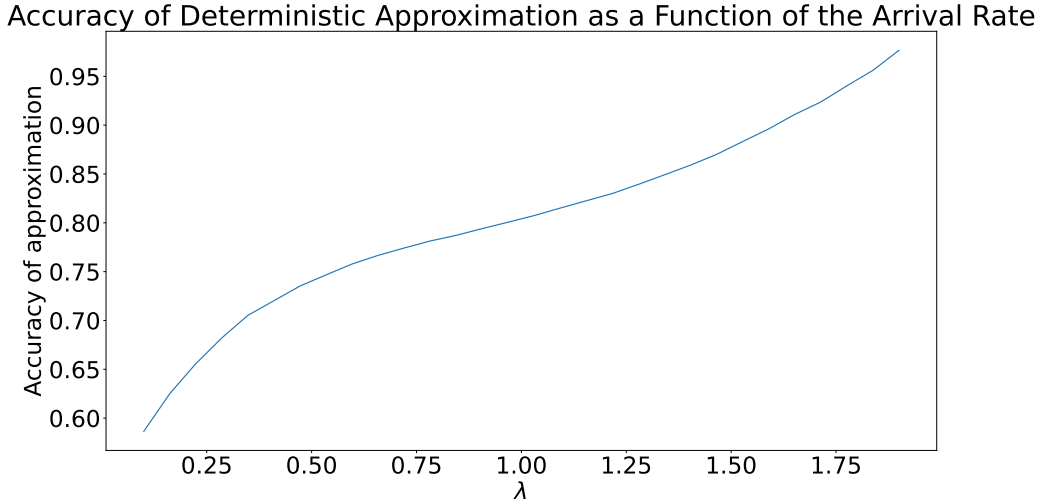


Figure 7.10: The higher the system load, the more accurate the deterministic approximation is.

7.1.3. Social Cost

In this subsection, the performance of the different strategies developed for the simplified network is analysed using a simulation. Simulations are run for the following strategies: weighted join shortest queue on servers S_1 and S_2 , the deterministic Nash approximation, the no starvation Nash approximation and the exact Nash equilibrium strategy. For each strategy, 25 simulations are run, each running for 500 000 time units, with a warm up time of 10 000. For each of the simulations, an average waiting time is recorded. The average waiting time serves as a measure of social cost, since it quantifies how long *all* customers must wait on average. Based on the 25 values, a 95 % confidence interval for the expected waiting time is computed.

The results are shown in Table 7.1 when all the service rates are equal to 1, and the arrival rate is equal to 1.6. The table shows that the value of information is large, as the average waiting time increases by a factor of 1.6 when using the unobservable Nash strategy as opposed to using the observable Nash strategy. Of the observable

strategies, the weighted join shortest queue performs worst. This is to be expected, as this strategy uses no queue length information of the second stage. The second worst observable strategy is the deterministic Nash approximation. The observable Nash strategy performs the best, with the no starvation approximation following closely.

Strategy	Average waiting time	95% CI of $\mathbb{E}(W)$
Unobservable Nash	6.778	[6.746, 6.810]
WJSQ	4.752	[4.732, 4.773]
Deterministic	4.546	[4.531, 4.561]
No Starvation	4.375	[4.358, 4.391]
Nash	4.319	[4.303, 4.336]

Table 7.1.: 95% confidence interval for the expected waiting time for each strategy when $\lambda = 1.6, \mu_i = 1, \forall i \in \{1, 2, 3, 4\}$

7.2. The Full Network

In this section, the numerical analysis of the strategies regarding the full network is discussed. We show how the approximate Nash strategy behaves in different scenarios, and how well it performs in a simulation, when compared to the WJSQ and the unobservable Nash strategy.

7.2.1. The Approximate Nash Strategy

In this subsection, the approximate Nash strategy is plotted for various values of the network parameters and the initial queue lengths. Figure 7.1 shows the approximate Nash strategy when the arrival rate and all the service rates are equal to 1 and both first stage stations are empty. The number of people in queues 3, 4 and 5, namely n_3, n_4 and n_5 , each range between 0 and 5. For each combination of n_3, n_4 and n_5 , a single point is plotted. If the approximate Nash strategy states a customer should go to queue 2 when observing the queue lengths $(n_1, n_2, n_3, n_4, n_5)$ upon joining the system, the point is coloured red. Otherwise, it is coloured blue. For each fixed value of n_3 and n_4 , the Nash strategy is to go to S_2 (S_1) when n_5 is small (large), as there are less people in the bottom (top) part of the network, so the Nash strategy suggests to go there. A similar symmetry can be observed when fixing n_5 along with n_4 . Thus, the Nash strategy behaves as we would expect.

Nash equilibrium strategy when $n_1 = 0$, $n_2 = 0$,
arrival rate is 1.0, service rates are all equal to 1.0

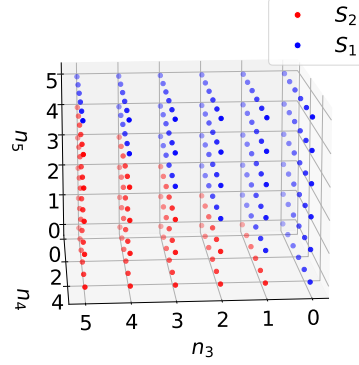


Figure 7.11: The Nash strategy behaves as we would expect in a symmetric setting.

A less intuitive behaviour can be seen in Figure 7.12. Here, the arrival rate and all service rates are equal to 1, with the exception of μ_1 which is set to 1.5. The first stage queues are again empty. One would expect to see more blue coloured points than in the previous figure, as server S_1 is faster. The plot, however, shows more red points. Take for example the diagonal part where $n_3 = n_5$. The points are mostly coloured red, meaning that the Nash strategy is to go to the slower server. There is no waiting in the first stage, so the cost is only the expected waiting time in the second stage. If a tagged customer goes to the faster server, they finish their first stage service more quickly (in expectation), and therefore have to wait longer for the second stage server. Hence, the Nash strategy is indeed to go to the slower server. The exception is when $n_4 = 0$. This is most likely due to the risk of overtaking: if, for example, the tagged customer sees queue lengths $(0, 0, 3, 0, 3)$ and they go to S_2 , there is a high probability they will use S_4 when done with their first stage service. However, there is also a non-negligible probability that another customer will join the system, join the faster server S_1 , and overtake them in queue 4.

Nash equilibrium strategy when $n_1 = 0$, $n_2 = 0$,
arrival rate is 1.0, $\mu_1=1.5$, other service rates equal to 1.0

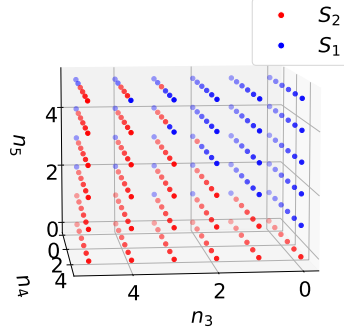


Figure 7.12: The Nash strategy behaves (seemingly) counter-intuitively due to our usage of the expected waiting time as a cost function.

The arguments above relied on the first stage queues being empty. If they are not, we observe again the intuitive behaviour of selfish customers using faster servers. Such a plot is shown in Figure 7.13, where again all rates are equal to 1 except μ_1 which is equal to 1.5, but now there are 3 customers in each first stage queue. The resulting behaviour is that S_1 is prescribed almost always. Only when queue 3 and queue 4 are busy and queue 5 is not, it is the best decision to go to S_2 .

Nash equilibrium strategy when $n_1 = 3$, $n_2 = 3$,
arrival rate is 1.0, $\mu_1=1.5$, other service rates equal to 1.0

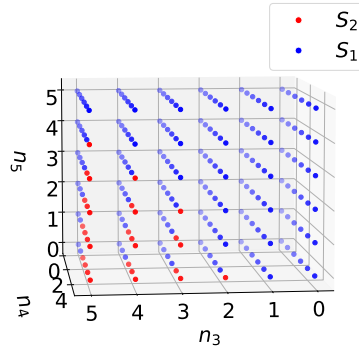


Figure 7.13: The Nash equilibrium strategy behaves as we would expect, sending more customers to a faster server.

7.2.2. Social Cost

In this subsection, the simulation performance of the Nash strategy approximation is studied. As for the simplified network, the average waiting time during a simulation

is taken as a measure of social cost. Three strategies are compared: the unobservable Nash strategy, the observable approximate Nash strategy and the weighted join shortest queue based on S_1 and S_2 .

Table 7.2 shows the average waiting times and the confidence intervals when all the service rates are equal to 1, and the arrival rate is equal to 1.6. As for the simplified network, the unobservable Nash strategy performs the worst in terms of the average waiting time, followed by the simple weighted join shortest queue approximation, which is followed by the approximate Nash strategy. Using the unobservable Nash strategy rather than the observable Nash strategy increases the waiting time by a factor of around 2.2, meaning that again the value of information is considerable. As seen in Chapter 5, in all instances studied, the added value of social planning was small. This is clear from the PoA being bounded by numbers only slightly greater than 1. Therefore, if a manager may either hire a central planner or give customers queue length information, our analysis suggests that they should opt for the latter.

Note also that the weighted join shortest queue strategy performs quite well - the average waiting time is only around 7% higher when using the weighted join shortest queue strategy as opposed to using the Nash strategy. This leads us to the following conjecture: if giving the customers queue length information at both stages is costly, it could be enough to provide only the first stage information. Note that providing only first stage information would not result in Nash being the same as WJSQ on S_1 and S_2 ; customers could still estimate how busy the second stage servers are. However, the Nash strategy probably would not be too far off from WJSQ on S_1 and S_2 . Hence, the Nash strategy would in that case still perform quite well.

Strategy	Average waiting time	95% CI
Unobservable	5.155	[5.135, 5.175]
WJSQ	2.382	[2.372, 2.392]
Nash	2.365	[2.357, 2.374]

Table 7.2.: 95% confidence interval for the expected waiting time for each strategy when $\lambda = 1.6, \mu_i = 1, \forall i \in \{1, 2, 3, 4\}$

8. Conclusion

In this paper, we studied two different routing strategies in a feed-forward, two stage queueing network. We considered two levels of information, the unobservable case where customers know only the network parameters, and the fully observable case where customers also know the queue lengths at each server. We took the expected waiting time as the cost function.

In the unobservable case, we numerically solved for the Nash equilibrium routing probabilities. We also formulated the social optimisation problem as a constrained optimisation problem, and solved the problem numerically to arrive at the socially optimal routing probabilities. We then studied the difference between the socially optimal and Nash routing probabilities, as well as the price of anarchy and the influence of the network parameters on the routing probabilities. Based on these numerical examples, the Nash equilibrium was found to be quite close to the social optimum. This indicates that, at least for the examples we studied, the added value of a social planner is small.

Next, we considered the fully observable problem, beginning with a related, but simplified version of the problem which is obtained by removing the middle server in the second stage. In doing so, we removed some of the difficulties that arise in a fully observable setting, such as complicated queue dependencies. This simplification allowed us to make various approximations to the Nash equilibrium, namely one that assumes service times are deterministic and one that assumes that second stage queues cannot become starved. We also solved for the exact Nash equilibrium in the simplified case by setting up a recursive relation in the expected sojourn time of a joining customer. The approximations were compared numerically to the exact Nash strategy under various network parameter values and queue length configurations. The performance of the aforementioned strategies was compared within a simulation, with the average waiting time in a simulation taken as a social cost metric.

This approach was used again in finding the Nash equilibrium for the fully observable case. A recursive relation was derived for the expected waiting time of a customer. The result was a system of equations, which becomes linear once an initial joining strategy for new arrivals is chosen. An algorithm similar to policy iteration was derived. The result was an approximate Nash strategy. The strategy is an approximation of the Nash strategy because in each iteration, we truncated the state space so that the system of equations became finite. The strategy was studied numerically for various network parameters and initial queue lengths. As in the simplified setting, its performance was compared to that of simpler strategies via a simulation. The Nash strategy was found to perform the best in terms of minimising average waiting time, followed somewhat closely by WJSQ. Both strategies outperformed the unobservable Nash equilibrium with a large difference in social cost, indicating that the value of information is considerable.

We were not able to derive the social optimum in the observable case. Obtaining this strategy could be one direction for future research. Additionally, a more extensive numerical study could be a direction for future research. Another direction could be to consider different network topologies. The topology we worked with was chosen since it appeared simple, and yet contained some interesting properties, such as branching paths that merge again in the second stage. It could be of interest to study different network topologies and see if similar effects are observed. For example, we could study if the unobservable Nash equilibrium and social optimum are close in other network topologies. Finally, only the most coarse and most fine levels of information were considered in this work - no information and full information. Other information levels could be considered, such as giving customers only the queue lengths at the stage they are about to join.

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A. Proofs

Lemma A.1. *Assume $p\lambda < \mu_3$, $\lambda(1-p) < \mu_5$, and $\lambda < \mu_4$. Then the functions $F_1, F_2 : [-1, 1] \mapsto \mathbb{R}^2$ given by $F_1(a, b) = (\mathbb{E}W_3 - \mathbb{E}W_4)(g(a, b))$ and $F_2(a, b) = (\mathbb{E}W_4 - \mathbb{E}W_5)(g(a, b))$ are continuous on $[-1, 1]^2$, and therefore $F = (F_1, F_2)^T$ is continuous on $[-1, 1]^2$.*

Proof. Since g is continuous everywhere, the discontinuities (a, b) of F_i are in a 1-to-1 correspondence with the discontinuities (p, q) of f through the relation $(a, b) = g^{-1}(p, q)$. Therefore, if we can argue that f has no discontinuities in $[0, 1]^2$ then it will follow that F has no discontinuities on $[-1, 1]^2$.

The discontinuities of f are the points where $\mathbb{E}W_3(q, r)$, $\mathbb{E}W_4(q, r)$ or $\mathbb{E}W_5(q, r)$ is not continuous. For the first expectation, these are (q, r) such that $\mu_3 = qp\lambda$. For the second one, they are (q, r) such that $\mu_4 = \lambda(p(1-q) + r(1-p))$. For the third expectation, it is points such that $\mu_5 = \lambda(1-r)(1-p)$.

We show that under the assumptions listed in the lemma, no element of $[0, 1]^2$ satisfies any of the three equations listed above. So let $(q, r) \in [0, 1]^2$. Then as $q < 1$, $pq\lambda < p\lambda < \mu_3$. As $r < 1$ and $1-q < 1$, $(p(1-q) + r(1-p))\lambda < (p+1-p)\lambda = \lambda < \mu_4$. As $1-r < 1$, $(1-r)(1-p)\lambda < (1-p)\lambda < \mu_5$.

Therefore f is continuous on $[0, 1]^2$, and F is continuous on $[-1, 1]^2$. $\square$

Lemma A.2. *Assume that $\frac{p\lambda}{\mu_3(\mu_3-p\lambda)} - \frac{(1-p)\lambda}{\mu_4(\mu_4-(1-p)\lambda)} > 0$ and that $\frac{p\lambda}{\mu_4(\mu_4-p\lambda)} - \frac{(1-p)\lambda}{\mu_5(\mu_5-(1-p)\lambda)} < 0$. Then the function F_1 is non-positive when $a = -1$, and non-negative when $a = 1$, and the function F_2 is non-positive when $b = -1$, and non-negative when $b = 1$.*

Proof. Let $r = \frac{b+1}{2} \in [0, 1]$ and $q = \frac{a+1}{2} \in [0, 1]$. We have

$$\begin{aligned} F_1(-1, b) &= f_1(g(-1, b)) = f_1(0, r) = \mathbb{E}W_3(0, r) - \mathbb{E}W_4(0, r) = 0 - \frac{\lambda(p+r(1-p))}{\mu_4(\mu_4-(p+r(1-p))\lambda)} \\ F_1(1, b) &= f_1(1, r) = \mathbb{E}W_3(1, r) - \mathbb{E}W_4(1, r) = \frac{p\lambda}{\mu_3(\mu_3-p\lambda)} - \frac{r(1-p)\lambda}{\mu_4(\mu_4-r(1-p)\lambda)} \\ F_2(a, -1) &= f_2(q, 0) = \mathbb{E}W_4(q, 0) - \mathbb{E}W_5(q, 0) = \frac{p(1-q)\lambda}{\mu_4(\mu_4-p(1-q)\lambda)} - \frac{(1-p)\lambda}{\mu_5(\mu_5-(1-p)\lambda)} \\ F_2(a, 1) &= \mathbb{E}W_4(q, 1) - \mathbb{E}W_5(q, 1) = \frac{(p(1-q) + (1-p))\lambda}{\mu_4(\mu_4-(p(1-q) + (1-p))\lambda)} - 0. \end{aligned}$$

It holds that $F_1(1, b) > 0, \forall b \in [-1, 1]$ if and only if $f_1(1, r) > 0, \forall r \in [0, 1]$, and as f_1 is decreasing in r , this happens if and only if $f_1(1, 1) > 0$. This is precisely what the

first assumption says. Similarly, $F_2(a, -1) < 0, \forall a \in [-1, 1]$ if and only if the second assumption is met.

Also, we can see from the first equation that $F_1(-1, b) < 0, \forall b \in [-1, 1]$ unless $p = 0$ and $r = 1$, which does not happen since we assumed all routes are used. By similar reasoning we have that $F_2(a, 1) > 0, \forall a \in [-1, 1]$. $\square$

B. Simulation

This simulation models a queuing network where customers arrive, select a queue for service, receive service, select a second stage queue for service and obtain service there before eventually exiting the system. As input, the user gives the desired simulation time, warm up time (the presumed time until a steady state is reached and the program starts tracking metrics), network parameters (arrival rate and service rates) and the strategy customers use to choose the queue they join.

The system operates based on events. There are six types of events, an arrival and a service at each of the five servers. When created, an event is added to the event heap. The heap is a data structure with a priority function. The priority function determines the order in which events are processed. The priority function used here is the scheduled time of the event, so that events occur in chronological order.

Customers arrive according to a Poisson process, with inter-arrival times following an exponential distribution. Upon arrival, a customer decides between queue 1 and queue 2 based on the strategy which is given as input. If a server in the chosen queue is idle, service begins immediately; otherwise, the customer joins the queue. Service times are also exponentially distributed. Once service in the first queue is completed, the customer makes a second routing decision, selecting among the available stage two queues before receiving further service.

The simulation tracks various metrics, including total waiting times, queue lengths, and routing decisions. The simulation runs for a specified time, collecting data after an initial warm-up period. Results include average waiting times per queue, the maximum observed queue lengths, the state each customer saw upon arrival and the decision the customer made.

C. Tables

Term	Expression
T_0	$(\mu_5 - \mu_4)\lambda(p - 1)^2$
T_1	$\lambda^2 p^4 (\mu_3 - \mu_4)^2$
T_2	$2p^3 \lambda (2\mu_3 p \mu_4 + \mu_3^2 \mu_4 + \mu_4^3 - \mu_3^2 p \lambda - \mu_3^3 - \mu_3 \mu_4^2 - \lambda \mu_4^2 p)$
T_3	$\mu_3^2 p^4 \lambda^2 + p^2 \mu_3^4 + p^2 \mu_4^4 + \lambda^2 \mu_4^2 p^4 + 4\mu_3^3 p^3 \lambda + 2\mu_3 p^3 \mu_4^2 \lambda$ $+ 2(\mu_3 \mu_4 p)^2 - 2\mu_4^3 p^3 \lambda - 2\mu_3 p^4 \lambda^2 \mu_4 - 4\mu_3^2 p^3 \lambda \mu_4$
T_4	$2p^2 \mu_3^2 (p \mu_4 \lambda - p \mu_3 \lambda - \mu_3^2 - \mu_4^2)$
T_5	$\mu_3^4 p^2$
T_6	$p^2 \lambda (\mu_3 - \mu_4)$
T_7	$p(\lambda p (\mu_4 - \mu_3) - (\mu_4^2 + \mu_3^2))$
T_8	$\mu_3^2 p$
T_9	$p \lambda (p - 1) (\mu_4 - \mu_3)$
T_{10}	$(p - 1) \mu_3^2$
T_{11}	$\mu_5 (p - 1) (\lambda - 2\lambda p - \mu_5) - \mu_4 (p - 1) (\lambda - 2\lambda p + \mu_4)$
T_{12}	$p \lambda (p - 1) (\mu_5 - \mu_4)$
T_{13}	$p^2 \lambda^2 \mu_4^2 + \mu_3^2 p^2 \lambda^2 + (\mu_3 p^2 \lambda)^2 + (p^2 \lambda \mu_4)^2 + 4\mu_3 \mu_4 p^3 \lambda^2$ $- 2\mu_3^2 p^3 \lambda^2 - 2p^2 \lambda^2 \mu_3 \mu_4 - 2p^4 \lambda^2 \mu_3 \mu_4 - 2p^3 \lambda^2 \mu_4^2$
T_{14}	$2p^3 \lambda \mu_3^2 \mu_4 + 2p \lambda \mu_3^2 \mu_4 + 4\mu_3^3 p^2 \lambda - 2p^3 \mu_3^3 \lambda$ $- 4p^2 \mu_3^2 \lambda \mu_4 - 2\mu_3^3 p \lambda$
T_{15}	$\mu_3^4 (p - 1)^2$
T_{16}	$p(\lambda(p - 1)(\mu_4 - \mu_5) - \mu_5^2)$
T_{17}	$p \mu_5 (\mu_5 + \lambda(p - 1)) + (p - 1) \mu_4 (\mu_4 - p \lambda)$
C_0	$T_5 T_0 + T_8 T_{10} T_{11} + T_{15} T_{17}$
C_1	$T_4 T_0 + T_7 T_{10} T_{11} + T_8 T_9 T_{11} + T_8 T_{10} T_{12} + T_{14} T_{17} + T_{15} T_{16}$
C_2	$T_3 T_0 + T_6 T_{10} T_{11} + T_7 T_9 T_{11} + T_7 T_{10} T_{12} + T_8 T_9 T_{12} + T_{13} T_{17} + T_{14} T_{16}$
C_3	$T_2 T_0 + T_6 T_9 T_{11} + T_7 T_9 T_{12} + T_{13} T_{16}$
C_4	$T_1 T_0 + T_6 T_9 T_{12}$

Table C.1.: Table containing the expressions used in Lemma 4.6